When Do Local Income Taxes Cause Residential Sorting?

Evidence from Pennsylvania's Piecewise EIT System

Matthew Hockert*

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Abstract

Do local income taxes cause residential sorting across sub-county boundaries? I use LEHD Origin-Destination data linking each worker's residence block to their workplace block at annual frequency, covering approximately 2,500 Pennsylvania municipalities and 500 school districts over 2002–2019.

Pennsylvania's Earned Income Tax operates a piecewise max-rule: each worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, where τ^{home} stacks the municipal and school-district resident rates and $\tau^{\text{nr,work}}$ is the work-location nonresident rate. A statutory rate change only moves the bill when the changed side of the max binds. I exploit this structure with a continuous stacked difference-in-differences design that decomposes each rate event by direction and binding-state transition, generating a within-design placebo: rate hikes whose post-event home stack stays below the nonresident rate (Inc-nobind) do not change the bill and should not move residents.

The preferred causal estimate is at the census block, where both rate-setting jurisdictions enter at their native geography. A 1 percentage-point municipal resident-rate

*University of Minnesota. Email: hocke069@umn.edu

increase in the binding cell reduces the log count of workers living in the block by -0.21 log points (approximately 19%), with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.213$, Wald $p = 0.003$). The Inc-nobind within-design placebo is statistically null on the same count outcome.

The municipality-level Inc-bind coefficient is larger (-0.57 log points on residents working in their own municipality) but pre-trended; I report it as policy-jurisdiction magnitude calibration rather than as the causal headline. School-district rate changes do not produce robust evidence of sorting and are treated as an aggregation diagnostic. Industry decomposition shows consumer-facing jobs fall in the binding cell while tradeable jobs are insulated, consistent with a local-services demand spillover from departing residents rather than direct nonresident-tax incidence.

1 Introduction

The Tiebout hypothesis predicts that residents sort across jurisdictions which offer different fiscal amenities and different costs and residents will 'vote with their feet' and choose specific bundles of amenities for a given amount of taxation, called the Tiebout margin (Tiebout, 1956). A survey of the literature conducted by Agrawal (2026) highlights the current findings of state and local income taxation and identifies sub-state (county, municipality, school districts) income-tax sorting as one of the field's open questions. The literature has focused mostly on state tax adjustments on high-earners (Agrawal & Foremny, 2019; Young et al., 2016). The research shows that top-income households are generally mobile when rates between states are large, and Agrawal (2015) documents sharp discontinuities among commodity-taxes at sub-state borders, but does not estimate the sub-state Tiebout margin for income taxes. The closest existing research comes from Yang and Heim (2017) which finds small migration effects for county-level income tax adjustments in Indiana.

This paper estimates the municipal and school district level migration effects among workers in Pennsylvania from 2002–2019, by merging local income tax data with the LEHD Origin-Destination Employment Statistics (LODES), which links each worker's residence to their workplace at annual frequency and at sub-municipal geography. In Pennsylvania there are roughly 2,500 municipalities and 500 school districts (SD) that set their own Earned Income Tax (EIT) rates (Agrawal, 2026; Bagchi, 2022), producing a deep cross-section of multiple sub-state income-tax opportunities. The EIT offers two additional dimensions of variation that my empirical design exploits. First, the EIT is set independently by two overlapping rate-setting jurisdictions (municipalities and school districts) and a resident's home-side tax bill is the sum of the municipal resident rate and the school-district rate. This paper's primary focus is the municipal resident rate while the school district rate enters the empirical design but is treated as secondary. Second, the EIT operates using a piecewise maximization rule where each worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, where $\tau^{\text{home}} = \tau^{\text{res,muni}} + \tau^{\text{SD}}$ is the sum of the municipal resident rate and the school-district resi-

dent rate at the worker's address (the home stack), and $\tau^{\text{nr,work}}$ is the nonresident rate at the work location. A statutory home-side rate change only moves the bill if the home stack is on the binding side of the max. Act 32 of 2008 (effective 2012) consolidated EIT collection into county-based Tax Collection Districts and mandated employer withholding at the higher of the two rates, which made the max-rule mechanically enforced. Bagchi (2022) estimates that Act 32 resulted in a 14% increase in compliance and much higher local revenues. My paper does not analyze compliance but rather estimates residential-sorting and provides additional analysis pre-post Act 32.

Beyond contributing to the general literature on local tax rate changes and residential sorting this paper's methodological contribution is utilizing the piecewise structure of the EIT to generate a binding-state decomposition to test how rate changes that move the tax bill from the non-resident side to resident side and producing a within-design placebo. A statutory rate change moves the worker's bill only when the home stack is on the binding side of the maximization function, but that same statutory change produces a predicted-null when the home stack stays below the work-side nonresident rate. Splitting rate-change events by whether or not they are binding isolates whose bill actually moves and those whose bill does not. Splitting rate-change events by binding state isolates the rate hikes that actually move the worker's bill and this within-design heterogeneity is what Agrawal (2026) identifies as missing from the existing U.S. local-income-tax literature. The decomposition is implemented at three units of observation (municipality, school district, census block) in a continuous stacked difference-in-differences design that uses never-changed municipalities as the control pool. Section 3 formalizes the binding-state cell decomposition and lays out the worker-bill differencing function each cell implies.

At the block level, the preferred causal estimate, a 1 percentage-point municipal resident-rate increase in the Inc-bind cell reduces the log count of workers living in the block by -0.21 log points (approximately 19%), the Inc-nobind, the within-design placebo, is null. The municipality-level result is larger (-0.57 log points on live-and-work residents, approximately

33% per pp change). School-district rate changes produce positive effects indicating that people move into school districts with higher taxes but the result is not robust given the lack of pre-trends. I also produce industry decomposition which shows consumer-facing jobs fall in the binding cell while tradeable jobs are insulated possibly indicating that local-services demand declines from departing residents rather than nonresident-tax incidence on jobs.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting and Act 32. Section 3 introduces the six-cell binding-state decomposition. Section 4 describes the downloading and merging of the multiple data sources: LODES, the DCED rate panel, and the municipal and school-district controls. Section 5 explains the continuous stacked difference-in-differences design at the three estimating units. Section 6 reports the pooled benchmark, the block-level causal headline, the municipality-level, and the industry mechanism. Section 7 reports robustness diagnostics, and Section 8 concludes.

2 Institutional Background

Pennsylvania has a deep cross-section of jurisdictions setting their own income tax rate: approximately 2,500 municipalities and 500 school districts each levy a separate Earned Income Tax on essentially all earned income earned in or by Pennsylvania residents (Agrawal, 2026; Bagchi, 2022). Unlike property taxes, which are paid by owners and capitalized into home values, the EIT falls directly on workers and varies across municipal boundaries, creating sharp spatial discontinuities in tax burden that can influence where workers choose to live and work.

2.1 Pennsylvania Earned Income Tax

Pennsylvania's EIT is mandatory for all municipalities under the Local Tax Enabling Act of 1965 (Act 511; Pennsylvania General Assembly, 1965). It applies to all earned income with no minimum threshold. The standard combined cap is 1%, typically split as 0.5% municipal

and 0.5% school district. Home Rule municipalities may exceed this cap (Pennsylvania General Assembly, 1972), and Philadelphia operates separately under the Sterling Act of 1932 (Pennsylvania General Assembly, 1932) at approximately 3.9%. EIT is a valuable revenue funder for local governments next to property-taxes. However, quantifying the effect of property tax millage on sorting would require a 2002–2019 municipality-level millage panel that is not currently available from DCED, and I leave the comparison to property-tax sorting as a separate exercise. Across 2002–2019, the panel contains 210 municipality resident-rate increases, 119 municipality decreases, 611 school-district increases, and 52 school-district decreases; the binding-state breakdown of these events is in Table 6.

Each Pennsylvania municipality has three EIT rates relevant to a worker: the municipality resident rate $\tau^{\text{res,muni}}$ levied on residents of the municipality, the municipality nonresident rate $\tau^{\text{nr,muni}}$ levied on workers employed in the municipality who live elsewhere, and the school-district rate τ^{SD} levied on residents only (school districts cannot tax nonresidents). Each address is associated with a six-digit Political Subdivision (PSD) code identifying its (municipality, school district) pair. Workers report this code on the Residency Certification Form, and employers use it to determine withholding obligations.

Under Act 32 (effective January 1, 2012), a worker's total EIT bill is determined by the maximum of the home-side rate stack and the work-location nonresident rate, and the bill is split piecewise between home and work jurisdictions:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr,muni}},\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr,muni}} \\ \left(0,\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr,muni}} \end{cases} \tag{1}$$

where j is the worker's home municipality and i is the work municipality. The work jurisdiction always collects its full nonresident rate first, then the home municipality and school district split whatever portion of the worker's bill is left over after the work claim. If the work nonresident rate exceeds the home stack, the home jurisdictions get nothing. The worker's

total bill is the higher of the two stacks:

$$\text{Bill}_{ji} = \max\big(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}}\big). \tag{2}$$

Equation (1) implies which margin a worker can use to respond to a rate increase. When the home stack binds ($\tau_j^{\text{home}} > \tau_i^{\text{nr,muni}}$), the worker can lower the bill only by moving home to a lower-rate municipality or school district. When the work nonresident rate binds, the worker can lower the bill only by changing jobs to a lower-nonres-rate work location, while moving homes without changing jobs has no effect. For workers who live and work in the same municipality, only the home stack applies($\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$).

A worker who lives in municipality j and moves to a lower-rate municipality j' while keeping the same job at i saves at least the SD rate at j: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which is at most the old bill τ_j^{home} minus τ_j^{SD}. The empirical design follows this logic directly: both the treatment $\Delta\tau^{\text{muni}}$ and the binding-state classification are built from the worker's home stack. The regressions therefore measure sorting in response to the same tax a mover would actually face.

Philadelphia is the one exception to the Act 32 framework. It operates under the Sterling Act of 1932 where Philadelphia residents pay the city resident rate regardless of work location with no offset and Philadelphia nonresidents pay the city nonresident rate, with no residual flowing to home jurisdictions in other Pennsylvania municipalities. My main estimates retain Philadelphia in the sample for completeness and the results are robust to dropping it.

Act 32 transformed the system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT mechanically binding where the rate changes now appear in every worker's paycheck through employer withholding, regardless of whether the worker resides in the taxing municipality. Second, it eliminated the

possibility of avoidance of the work-location nonresident claim through weak enforcement. After Act 32, a worker could no longer effectively duck the nonresident portion owed to the work municipality.

However, Act 32 did not eliminate all tax savings from residential relocation. The piecewise rule in Equation (1) continues to imply that a worker whose original home stack binds can reduce the bill by moving to a sufficiently lower-stack home jurisdiction where the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which falls below the old home stack τ_j^{home} whenever $\tau_{j'}^{\text{home}} < \tau_j^{\text{home}}$ and the original home stack was the binding side. Act 32 thus changed enforcement and made the max-rule mechanically binding through withholding; it did not eliminate the residential-relocation margin that the mechanism in this paper exploits. Bagchi (2022) documents the quantitative consequences of Act 32 directly: consolidated employer withholding increased PA EIT compliance, raised reported earned income in the rate-raising municipality, and shifted the de facto incidence toward workers. The present paper complements that work by isolating the residential-sorting response to the same withholding-enforcement regime, exploiting the cross-sectional variation in whether a given rate change actually moves the worker's bill (the binding-state decomposition); the pre/post-Act-32 split in Section 7 shows the binding-cell sorting response strengthens (the effect existed prior to Act 32) after Act 32, complementing the compliance-response finding.

School districts in Pennsylvania also levy an EIT under Act 511 (Pennsylvania General Assembly, 1965), typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (Pennsylvania General Assembly, 2006) (the Taxpayer Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is independent of municipal fiscal decisions. I find that the correlation between the municipal and school-district resident rates is only weakly negative of -0.12 indicating the two taxing jurisidction move on reletively separate schedules. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change

independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

3 Conceptual Framework: Binding-State Decomposition

The piecewise tax allocation in Equation (1) is the causal identification backbone of my empirical design. A worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only moves the bill when its side of the max binds. The implied change in the worker's bill from a single rate event is

$$\Delta T \;=\; \max(\tau^{\text{home}}_{\text{post}}, \tau^{\text{nr}}) - \max(\tau^{\text{home}}_{\text{pre}}, \tau^{\text{nr}}),$$

which depends jointly on the direction of the rate change and on which side of the max is binding pre- and post-event. This tax difference from pre and post rate changes can produce a strong behavioral response, a moderate response, or no response at all depending on the binding-state transition.

3.1 Six Cells per Rate-Setter

Two definitions structure the decomposition. First, the piecewise rule $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$ has two sides: the home side $\tau^{\text{home}} = \tau^{\text{res,muni}} + \tau^{\text{SD}}$, set at the worker's residence, and the work side $\tau^{\text{nr,work}}$, set at the worker's work municipality. Whichever side is larger is the binding side and determines the bill. Second, three rate-setters adjust statutory rates independently across municipality-years: the municipality resident rate $\tau^{\text{res,muni}}$, the school-district rate τ^{SD}, and the municipality nonresident rate $\tau^{\text{nr,muni}}$. The decomposition is built one rate-setter at a time.

For each rate-setter τ, a cell classifies a municipality by (i) the direction of its first rate event in τ (Increase or Decrease) and (ii) the binding-state transition of its home stack across

that event: stayed-binding, flipped, or stayed-non-binding. The cross gives six cells plus a seventh never-changed control pool (Table 1). Each municipality therefore carries three cell labels (one for each of $\tau^{\text{res,muni}}$, τ^{SD}, and $\tau^{\text{nr,muni}}$) and the binding state for each is determined by where its home stack sits relative to its nonresident rate at the time of that event.

Cell	Direction	Pre-binding?	Post-binding?	Bill effect
Inc-bind	$\uparrow$	yes	yes	rises by full $\Delta\tau$
Inc-becomes	$\uparrow$	no	yes	rises by part of $\Delta\tau$
Inc-nobind	$\uparrow$	no	no	no change (placebo)
Dec-bind	$\downarrow$	yes	yes	falls by full $\Delta\tau$
Dec-becomes-non	$\downarrow$	yes	no	falls only to τ^{nr}
Dec-nobind	$\downarrow$	no	no	no change (placebo)
NC	none	—	—	control pool

Table 1: Binding-state cells per rate-setter. For Inc-becomes the bill rises only by the portion of $\Delta\tau$ above the nonresident rate; for Dec-becomes-non the bill falls only until the home stack hits the nonresident floor.

The fundamental testable prediction is an ordered ranking of cell magnitudes where Inc-bind effects exceed Inc-becomes effects, which in turn exceed Inc-nobind effects, with the last close to zero and serving as the within-design placebo. A tax salience argument would state that workers in the Inc-becomes cell experience a discrete switch from paying zero home-side tax to paying a positive amount, which should trigger a large behavioral response. The central empirical comparison of the paper is between Inc-bind and Inc-nobind where if statutory rates matter mechanically through tax liability, the residential response should appear in Inc-bind and not in Inc-nobind. If Inc-nobind moves, the mechanism interpretation fails and the response is something other than the binding rule because Inc-nobind events do not change the worker's bill. This makes the design self-falsifying in a way that a conventional pooled Inc/Dec specification is not. In the pooled specification, every increase is a treated event, while here the institutional rule predicts that only a subset of statutory increases

should generate a response.

4 Data

4.1 Data Sources

The analysis merges the LEHD residence–workplace dataset and the Pennsylvania EIT rate records to create a panel of rate changes and workforce data over space and time. The two datasets do not share common identifiers, and was bridged using TIGER 2020 block geometry, which nests blocks within county subdivision code(cousubs). A municipality-name crosswalk maps each DCED municipality ID to its corresponding COUSUBFP code, matching 2,496 of 2,505 PA municipalities (99.6%). The remaining 9 are small boroughs that have either dissolved or merged during 2002–2019 and are dropped from the analysis.

The residence–workplace data come from LEHD LODES Version 8, 2002–2019 (U.S. Census Bureau, 2024b). The Census Bureau applies synthetic noise infusion at the block level to protect respondent confidentiality and is calibrated so that aggregation to tract and larger geographies approximates ground truth (Graham et al., n.d.). The sample period of 2002–2019 reflects LODES coverage beginning in 2002 and 2020–2023 to not include disruptions in the data resulting from the COVID Pandemic.

Municipal and school-district EIT rates come from the Pennsylvania Department of Community and Economic Development (DCED) MunStats portal (Pennsylvania Department of Community and Economic Development, 2024a), scraped annually. In some DCED reports, the flat-dollar Local Services Tax is filled in the same field as the EIT in some years, so rates above 5% are treated as data-entry errors and dropped except for Home Rule cities and Philadelphia.

Municipal fiscal controls come from the PA DCED Statewide Municipal Annual Financial Report database (Pennsylvania Department of Community and Economic Development, 2024b). School-district fiscal controls come from the Census Annual Survey of School System

Finances (F-33) (U.S. Census Bureau, 2024a); the NCES `LEAID`-to-DCED `sd_id` crosswalk covers all 500 school districts.

4.2 Descriptive Statistics

This section presents the summary statistics across the three aggregation levels the empirical strategy estimates. For the block level, the primary outcome variable is the log count of workers living in the block. The count directly measures the residential location of active workers and is less sensitive in smaller geographies compared to shares which are used at higher level aggregations like the municipality or school district and I report share outcomes as supporting evidence rather than as the primary estimand. At those higher aggregations I also report resident-worker counts, workplace counts, in-commuter flows. Fiscal outcomes used as control variables (log EIT revenue, log total tax revenue, log total revenue, log police expenditure) are reported in Appendix C.

Tables 3–4 summarize the three estimation panels.

Table 2: Block-year summary statistics (2002–2019)

Variable	N	Mean	SD	Min	Max
Resident rate τ^{M} (%)	4,782,928	0.762	0.799	0.000	4.331
Nonresident rate $\tau^{nr,M}$ (%)	4,181,214	1.063	0.657	0.000	3.820
SD rate τ^{SD} (%)	4,782,928	0.607	0.376	0.000	2.050
Home count (workers living)	5,030,958	18.822	31.677	0.000	1572.000
log home count	4,795,213	2.293	1.198	0.000	7.360
Work count (jobs in block)	5,030,958	18.826	138.957	0.000	20945.000
log work count	1,894,432	2.256	1.663	0.000	9.950
Share live-and-work in muni	4,795,213	0.154	0.222	0.000	1.000
Share work outside muni	4,795,213	0.846	0.222	0.000	1.000
Blocks per muni	2,561	120.922	384.585	1.000	16550.000

Notes: Block-year level; one observation per (census block, year). Panel covers 309,680 blocks in 2,561 municipalities.

Table 2 presents the summary statistics of the block panel outcomes. The block panel is saturated with observations with roughly 5,030,958 block-years across 309,680 census blocks and 2,561 municipalities. The mean resident rate is roughly 0.76% with an SD rate of 0.80pp, where Philadelphia, Pittsburgh, and the other Home-Rule cities drive up a disproportionate share of PA's blocks pulling the block-weighted distribution toward their higher rates. The same logic explains why the block-level nonresident-rate mean (1.06%) much higher as the cities with elevated home-side rates also tend to have higher nonresident rates. Nearly 850,000 block-years are missing a nonresident rate (the row has 4.18M obs vs. 4.78M for the other rates). The outcome variables show that the typical block has $\approx$ 19 workers living in it (max 1,572) and $\approx$ 19 jobs in it (max 20,945 and SD of 139). The raw-count means between the log home count and log work count are nearly identical but are dispersions between the

two are quite different with home counts have a standard deviation of 32 on a mean of 19, while work counts have a standard deviation of 139 on the same mean. This shows that residential workers tend to spread fairly uniformly across blocks, but jobs concentrate in a small number of commercial blocks. The mean share of in-block residents who also work in the same muni is 15%, but most blocks still send the bulk of their workers out of the muni (84.6% on average), indicating that residential sorting in PA operates on a thin within-muni margin even at the block level.

Table 3: Muni-year summary statistics (2002–2019)

Variable	N	Mean	SD	Min	Max
Resident rate τ^M (%)	43,374	0.525	0.163	0.000	4.331
Nonresident rate $\tau^{nr,M}$ (%)	34,080	0.883	0.237	0.000	3.820
SD rate τ^{SD} (%)	43,374	0.622	0.289	0.000	2.050
Home stack $\tau^M + \tau^{SD}$ (%)	43,374	1.147	0.316	0.000	4.331
log live-and-work	40,748	3.638	2.011	0.000	12.877
log total jobs	45,291	5.819	2.006	0.000	13.353
Share live-and-work in muni	45,291	0.095	0.081	0.000	1.000
Total jobs	45,291	2076.691	13031.730	1.000	629698.000

Notes: Muni-year level; one observation per (municipality, year). Rates in percentage points. $\tau^{nr,M}$ has reporting gaps before Act 32; the N for that row is smaller than the full panel.

Table 3 presents the municipality level summary statistics. As mentioned above the outcome variables are slightly different than the block level because of the difficultly to use shares at a lower level like census blocks. Table 3 shows that there are rougly 43,374 muni-year observations, and the resident rate is tightly clustered near the Act 511 default at $\bar{\tau}^M = 0.53\%$ and a standard deviation of 0.16pp. Like the block level, there is a long right tail indicated by the max rate of 4.33% concentrated in Philadelphia and a handful of

Home-Rule and Act 47 distressed cities. The block-level tax-rate distributions look different from the muni-year averages potentially due to larger munis contribute proportionally more blocks. The nonresident rate has 34,080 observation, about 9,300 fewer observations than the resident rate, due to DCED's lack of coverage for the nonresident-rate field being incomplete before Act 32 mandated reporting. The SD rate has nearly twice the muni-rate dispersion (SD 0.29pp vs. 0.16pp) and maxes at 2.05%, potentially making the school-district rate-setter a larger statistical lever. The home stack averages 1.15%, above the 1% Act 511 cap, because Act 1 SD increases and Home-Rule muni rates push the upper level past the ceiling. Outcome variables show that total jobs span 1 to 629,698 with an SD of 13,032 on a mean of 2,077, indicating why the regressions use logs. And only 9.5% of muni jobs are held by residents of the same muni showing that residential sorting operates on a thin within-muni margin.

Table 4: School-district-year summary statistics (2002–2019)

Variable	N	Mean	SD	Min	Max
SD rate τ^{SD} (%)	49,226	0.610	0.278	0.000	2.050
Resident rate τ^{M} (%) (muni-year)	49,226	0.523	0.162	0.000	4.564
Nonres rate $\tau^{nr,M}$ (%) (muni-year)	38,536	0.881	0.237	0.000	3.967
share_state_rev_sd	9,278	0.416	0.166	0.000	0.821
share_local_rev_sd	9,278	0.536	0.186	0.103	1.000
share_federal_rev_sd	9,278	0.048	0.033	0.000	0.247
share_property_tax_sd	9,278	0.420	0.176	0.000	0.853
share_instructional_sd	9,278	0.522	0.070	0.000	1.000
share_support_sd	9,278	0.279	0.039	0.000	0.438

Notes: SD-year level. F-33 share controls are constructed from Census F-33 revenue/expenditure totals.

Table 4 shows the school district descritpive statistics. I aggregated rates to appear at the muni-year level because each muni inherits its school district's rate. At this level there are roughly 49,226 observation while the fiscal shares have 9,278 observations because they are reported at the school district-year level with roughly 500 districts by 18 years. The school district rate averages 0.61% with SD 0.28pp and caps at the Act 1 ceiling of 2.05% which drives up school district rates more aggressively than munis after 2006, and generates SD-side identification. PA school districts get 54% of revenue from local sources on average and 42% from state aid, and property taxes are about 78% of local revenue indicating that an Act 1 EIT increase is a one-for-one substitution against the largest local revenue source meaning it is a change in tax composition rather than tax burden.

Table 5: Cell counts: muni resident-rate side, 2002–2019

Cell	Munis	Muni-years	Mean dose (pp)	Extreme dose (pp)
Never-changers	1,730	31,132	0.000	0.000
Inc-bind	24	432	0.168	0.850
Inc-becomes	43	774	0.246	1.250
Inc-nobind	20	360	0.294	0.650
Dec-bind	4	72	-0.138	-0.500
Dec-becomes-non	3	54	-0.278	-0.500
Dec-nobind	11	198	-0.380	-1.000

Notes: Cells classify each muni's first resident-rate event by direction (Inc/Dec) and the binding-state transition relative to the muni's nonresident rate. NC is the never-changed control pool. Dose is the cumulative $\Delta\tau$ within the 5-year event window. The Extreme dose column reports the max cumulative dose for Inc cells and the min (most negative) cumulative dose for Dec cells, since cell assignment is determined by the first event but subsequent reversals can pull the signed cumulative path back across zero.

Table 5 reports the binding-state cell assignments on the muni resident-rate side. Of $\sim$ 2,500 municipalities, 1,793 never change their resident rate and form the NC control pool;

only 194 are assigned to any treated cell, which is the thin tail that carries identification. The three increase cells split as 24 Inc-bind ($\bar{\Delta\tau} = 0.19$pp at $k = +5$, max 1.50pp), 43 Inc-becomes ($\bar{\Delta\tau} = 0.34$pp, max 4.33pp driven by Philadelphia), and 20 Inc-nobind ($\bar{\Delta\tau} = 0.27$pp). Inc-becomes carries a larger average dose than Inc-bind because crossing the binding threshold typically requires a bigger statutory move than ratcheting up an already-binding rate. The decrease cells are thin — 11 Dec-bind, 4 Dec-becomes-non, 15 Dec-nobind — and the Extreme dose column reports the most-negative point of the cumulative path in the 5-year window. Some Dec munis reverse direction within that window (an initial decrease later followed by an increase), so signed mean dose and signed extreme dose can have opposite signs by row; this is the empirical reason the headline focuses on the Inc cells and the flip-flopper-robustness check is run separately.

Figure 1 plots the empirical CDF of the three rate variables that enter the piecewise max-rule: the municipality resident rate τ^M, the municipality nonresident rate $\tau^{nr,M}$, and the home stack $\tau^M + \tau^{SD}$. Under Act 511 of 1965 (Pennsylvania General Assembly, 1965), the combined home-side EIT is capped at 1% for ordinary boroughs and townships and is conventionally split 0.5% municipal plus 0.5% school district; the vast majority of Pennsylvania municipalities sit at this statutory default and never adjust. Approximately 121 of the ∼2,500 municipalities change their resident rate at any point in the sample, and most school districts likewise hold at 0.5%. The ECDF makes this concentration visible: roughly 95% of municipality-years sit at or below the 0.5% resident-rate cap, and the home-stack CDF reaches $\sim 78\%$ by the 1% combined cap. The long right tails correspond to two policy carve-outs: Act 47 distressed cities (Pennsylvania General Assembly, 1987) (Pittsburgh-area municipalities, Wilkes-Barre, Hazleton, Johnstown, and others) that exceeded the municipality half of the 1% cap, and Act 1 school districts (Pennsylvania General Assembly, 2006) that converted property-tax authority into EIT authority. Philadelphia operates under the Sterling Act of 1932 (Pennsylvania General Assembly, 1932) rather than Act 511 and accordingly sits at the far right of the resident-rate distribution. These tail municipalities and

school districts generate the identifying variation that the binding-state decomposition then splits by whether each statutory rate change moved the worker's bill.

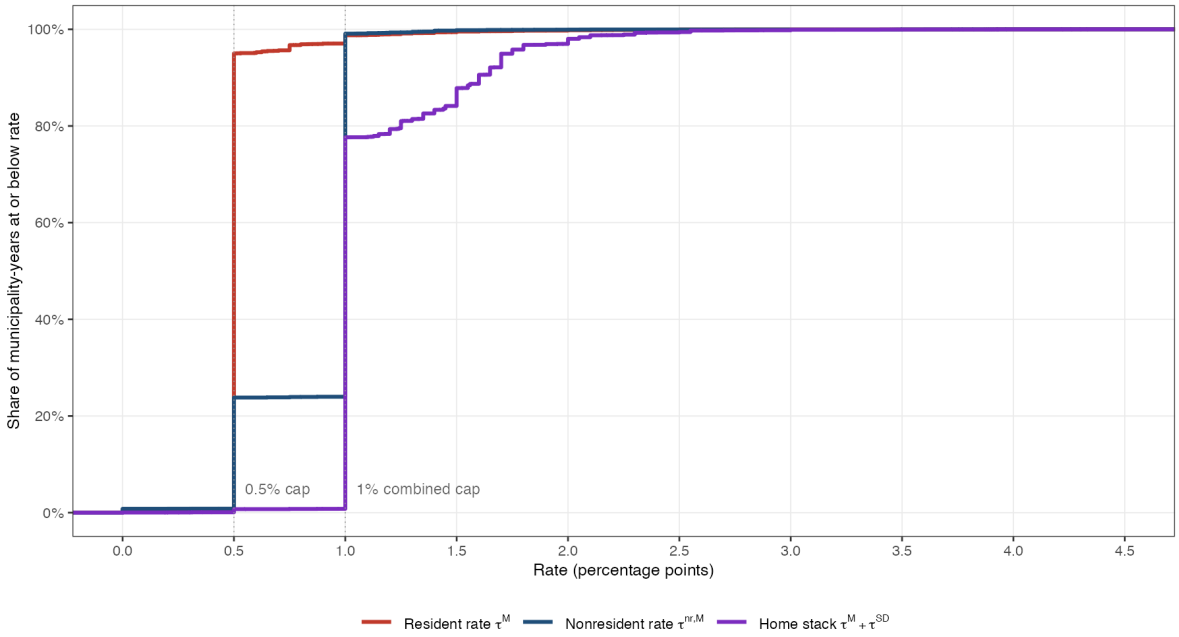


Figure 1: Empirical CDF of Pennsylvania EIT rates, 2002–2019. Each step shows the share of municipality-years at or below a given rate, separately for the municipality resident rate τ^{M} (red), the municipality nonresident rate $\tau^{nr,M}$ (blue), and the home stack $\tau^{M} + \tau^{SD}$ (purple). Dotted reference lines mark the 0.5% municipality cap and the 1% combined home-side cap under Act 511 of 1965. Approximately 95% of municipality-years sit at the 0.5% resident-rate cap and $\sim$ 78% at the 1% combined home-stack cap; the long right tails are Act 47 distressed cities, Home-Rule cities, and Act 1 school districts.

Figure 2 reports the timing of municipal resident-rate changes across the panel. Increases outnumber decreases nearly two-to-one overall (210 increases vs. 119 decreases over 2002–2019), and the gap widens further after Act 32 took effect in 2012: roughly 80% of post-2012 rate-change events are increases, consistent with consolidated withholding raising the political cost of cutting a rate that is now mechanically enforced.

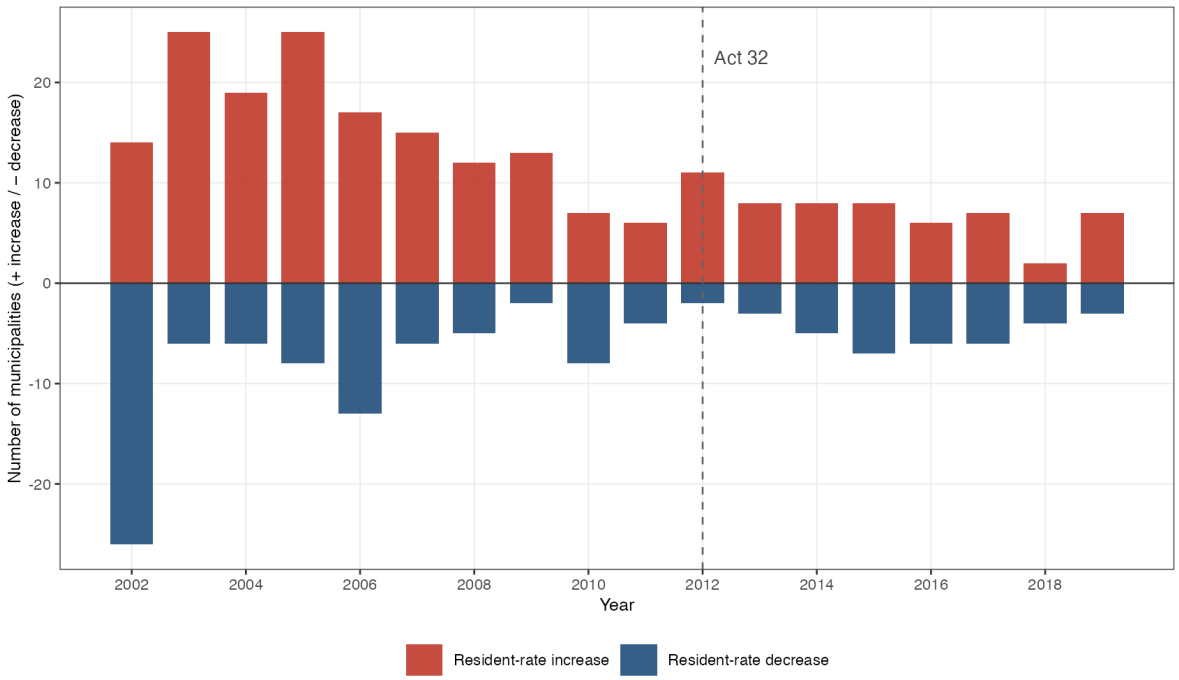


Figure 2: Number of PA municipalities changing their resident rate, by year. Increases shown above zero; decreases shown below. Dashed line: 2012, when Act 32 consolidated withholding.

Figure 3 documents the institutional break at Act 32. Before 2012, fewer than 60% of PA municipalities reported a nonresident rate to DCED; after 2012, when consolidated withholding made the max-rule mechanically binding, coverage rises sharply. This is the pre-/post-Act-32 enforcement break used in the heterogeneity test of Section 7.1.

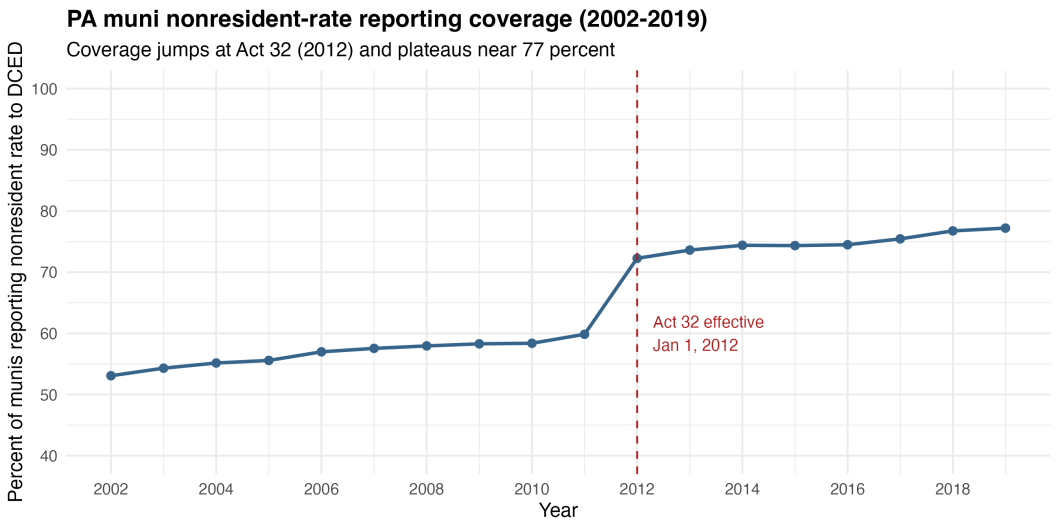


Figure 3: Share of PA municipalities reporting a nonresident rate to DCED, by year. The break at 2012 reflects the Act 32 requirement that employers withhold at the higher of the home resident and work nonresident rates.

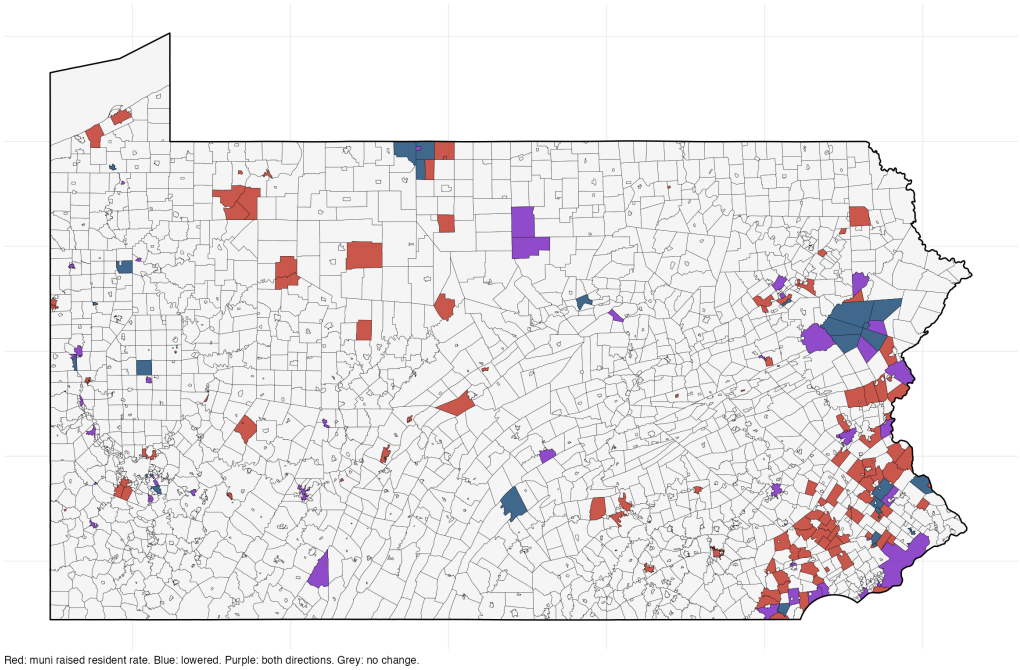


Figure 4: Pennsylvania EIT rate changes, 2000–2019. Red = increased; blue = decreased; purple = both; grey = no change.

4.2.1 Cell counts and stacked-design support

Before turning to the empirical strategy, this subsection documents the sample composition of the binding-state decomposition. The purpose is twofold: to verify that each cell in Section 5 has enough municipalities to identify a per-pp coefficient, and to characterize which types of Pennsylvania municipalities populate the treated cells—a within-state heterogeneity that matters because PA legal-organization classes (city, borough, township) have different revenue authorities and fiscal-capacity profiles.

Table 6 reports the number of unique municipalities in each binding-state cell on each rate-setter after the NC-purity filter described in Section 5.

Table 6: Cell composition by rate-setter (drop sample, post-NC-fix)

Side	Inc-bind	Inc-becomes	Inc-nobind	Dec-bind	Dec-becomes-non	Dec-nobind	NC
Muni resident (τ^{muni})	40	58	26	11	4	15	2,406
School district (τ^{SD})	75	197	14	4	10	—	2,253
Muni nonresident (τ^{nr})	0	0	171	0	5	109	2,276

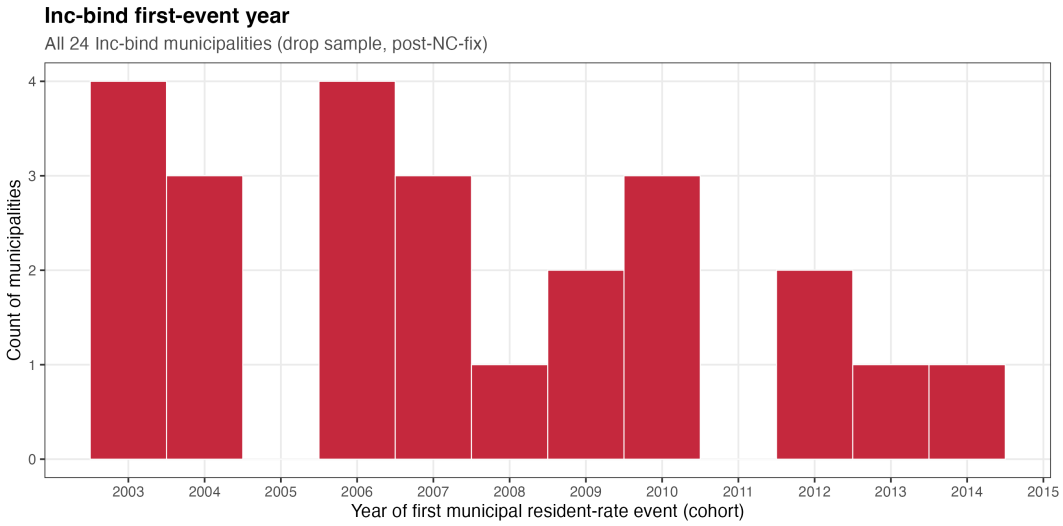
Notes: Cells classified at each municipality's first observed rate event on the relevant rate-setter. Munis with rate changes whose first-event binding state cannot be determined (NA pre-event nonresident rate) are dropped (not pooled with NC). On the nonresident side, Inc-bind and Inc-becomes cells are empty in PA because the municipality's nonresident rate almost never exceeds the municipality's own home stack at the municipality-self-pair level.

The 24 Inc-bind municipalities, when aggregated, are disproportionately concentrated in older industrial cities and boroughs of Pennsylvania (Wilkes-Barre, Coatesville, Aliquippa, Johnstown, Hazleton). Decrease cells are extremely thin (4 Dec-bind, 3 Dec-becomes-non, 11 Dec-nobind in total), so the binding-rule symmetry test between Inc and Dec is not feasible under conventional standard-error inference.

The stacked design's identifying support comes from the joint distribution of cohort timing and event-time coverage. Figure 5 reports the distribution of municipality-rate first-event years for the 24 Inc-bind municipalities. The cohort distribution is bimodal: an early

spike in 2003–2004 (boroughs and townships responding to baseline EIT setup) and a 2006 spike of 10 units (Act 1 of 2006 enabled SD-rate increases that mechanically flipped some municipalities into Inc-bind via the joint $\tau^{\text{muni}} + \tau^{\text{SD}}$ stack), with a long tail through 2014.

Figure 5: Inc-bind first-event year.



Notes: Counts of Inc-bind municipalities by year of first municipality-rate event, aggregated across municipality classifications. The 2006 spike reflects Act 1 of 2006 (school-district EIT authorization), which mechanically flipped some municipalities into the binding cell via the joint home-stack $\tau^{\text{muni}} + \tau^{\text{SD}}$.

5 Empirical Strategy

The empirical framework is a continuous stacked difference-in-differences design with binding-state cell decomposition, estimated on a panel spanning 2002–2019 at three units of observation: municipality, school district, and census block. Pre-2002 data are not consistently available in LODES; post-2019 years are excluded because COVID disrupted commuting patterns and ARPA fiscal transfers distorted municipal budgets. Each estimating unit plays a different role. The block-level estimate is the preferred causal specification because both rate-setting jurisdictions enter the regression at their native geography and the realized-dose pre-trend on the binding cell is clean. The municipality-level estimate is retained as policy-jurisdiction magnitude calibration, and the school-district aggregation is used as a

diagnostic. The first-difference dose-response specification estimates the post-event effect of a one percentage-point change in the relevant tax rate; a separate realized-dose event-study diagnostic tests for differential pre-event trajectories across units of varying eventual treatment intensity.

The resident-side specification includes two simultaneous rate-setter treatments: $\Delta\tau_{it}^{\text{muni}}$, the cumulative change in the municipality resident rate from baseline, and $\Delta\tau_{it}^{\text{SD}}$, the cumulative change in the school-district rate. Baseline for each is the rate at year $c_i^s - 1$, where c_i^s is unit i's first event year on rate-setter $s \in \{\text{municipality, SD}\}$, and cumulative dose accumulates with subsequent rate changes within i. The work-side nonresident rate τ^{nr} enters the binding-state classification of cells but is not itself a treatment in the headline regression: I focus on the resident-side mechanism and defer the work-side behavioral response to a separate analysis.

Each rate-change event is classified by direction $\in \{\text{Inc, Dec}\}$ and binding-state transition $\in \{\text{bind, becomes, nobind}\}$ to yield the six non-NC cells introduced in Section 3.1. For each unit i and each rate-setter s, I locate the first rate event year c_i^s and assign i to one of the six cells based on the direction of the change and the binding state of the home stack relative to $\tau^{\text{nr,muni}}$ in the pre- and post-event years. The NC reference category is restricted strictly to units whose rate τ^s never changes over 2002–2019. Units with rate changes whose first event has an undefined binding state—typically because the municipality's nonresident rate is missing in the pre-event year due to DCED reporting gaps before Act 32—are categorized as "Uncertain" and dropped: pooling them with NC would contaminate the control trend with treated-but-unclassified units, while reporting them in their own cell would mix true binding states with measurement gaps. Municipality-level regressions further impose a minimum employment threshold of 50 total jobs. The threshold protects the share-based outcomes (*share_live_in*, *share_work_outside*) from mechanical measurement noise: at N workers, a single worker moving changes the share by $1/N$, so at $N = 10$ a one-worker move changes the outcome by 10 percentage points, dominating the per-pp treatment magnitude. At $N = 50$

the one-worker noise floor is 2%, roughly an order of magnitude below the headline coefficient. The threshold also matches the LODES synthetic-noise-infusion design: the Census Bureau's noise calibration is validated at the tract level and above (Graham et al., n.d.), which in Pennsylvania typically contains at least 50 workers. Choosing exactly 50 rather than 20 or 100 is a judgment call; sensitivity analysis in Section 7 confirms the headline Inc-bind coefficient is stable across thresholds from 10 to 500. Imposing the threshold drops 185 of 2,559 PA municipalities (7.2%), almost all of which are sparsely populated rural townships in the NC pool; the treated cohort is essentially unaffected.

Outcomes are run in parallel across counts and shares. The municipality-level workplace and residence outcomes are the logs of `live_and_work` (residents employed in the municipality), `in_commuters` (workers in the municipality from elsewhere), `out_commuters`, `total_jobs`, and `total_residents`, plus the shares of workers living in their own municipality, of workers who are nonresidents, and of residents working outside the municipality. The school-district aggregation uses the same outcomes computed by summing block-level LODES counts up to the SD: each block has a unique cousub assignment (via the TIGER 2020 block-to-cousub crosswalk), each cousub has a primary SD assignment (the modal `sd_id` in the DCED rate panel; 97.6% of munis are single-SD), and the SD-level outcomes are block sums via this two-step block $\rightarrow$ cousub $\rightarrow$ SD path. At the block level, outcomes are the log count of workers living in the block, the log count of workers working in the block, the share of block residents working in their own municipality, and the share working outside; geometry and aggregation details are in Section 5.3.

Fiscal controls are share-based and time-varying, with the exact set keyed to the estimating unit. The shares are constructed as numerator-over-total ratios so that they absorb compositional changes in the municipality's or district's fiscal posture without conflating with the level of total spending.

Municipality-level regressions use six share controls from the DCED Annual Financial Report. Three are expenditure shares: `share_police` (police-services line / total expendi-

ture), share_streets (streets and highways), and share_debt_svc (debt service: principal and interest payments on outstanding general-obligation and revenue debt). Three are revenue shares: share_eit (Earned Income Tax revenue collected / total revenue), share_re (real-estate property tax revenue / total revenue), and share_intergov (intergovernmental revenue, which is grant and pass-through revenue from state and federal governments—e.g. Liquid Fuels Tax distributions, Community Development Block Grants, ARPA transfers—divided by total revenue).

School-district regressions use eight share controls from the Census Annual Survey of School System Finances (F-33) (U.S. Census Bureau, 2024a). Four are revenue shares: share_state_rev_sd (state aid: basic education subsidy, special-education subsidy, transportation reimbursement, etc.), share_local_rev_sd (locally raised revenue: property tax, EIT, occupation tax, public-utility taxes), share_federal_rev_sd (federal Title I, IDEA, school-lunch, and other direct or pass-through federal aid), and share_property_tax_sd (real-estate property-tax revenue as a share of total revenue, which is a subset of share_local_rev_sd and isolates the SD's primary local-tax lever). Four are expenditure shares: share_instructional_sd (teacher salaries, classroom supplies, instructional staff), share_support_sd (non-instructional operating costs: administration, transportation, maintenance, food service, student services), share_capital_sd (capital outlay—construction, renovation, equipment), and share_debt_int_sd (interest payments on the district's outstanding debt; principal appears as a capital outlay rather than as a current expense).

Block-level regressions use both sets of controls simultaneously, since each block is exposed to both rate-setters' fiscal posture. Figure 6 plots the empirical distribution of each share variable pooled across municipality-years (municipality shares) or SD-years (SD shares); cross-unit variation is broad on every dimension, so the share controls absorb meaningful fiscal heterogeneity without collinearity with the rate-treatment variable.

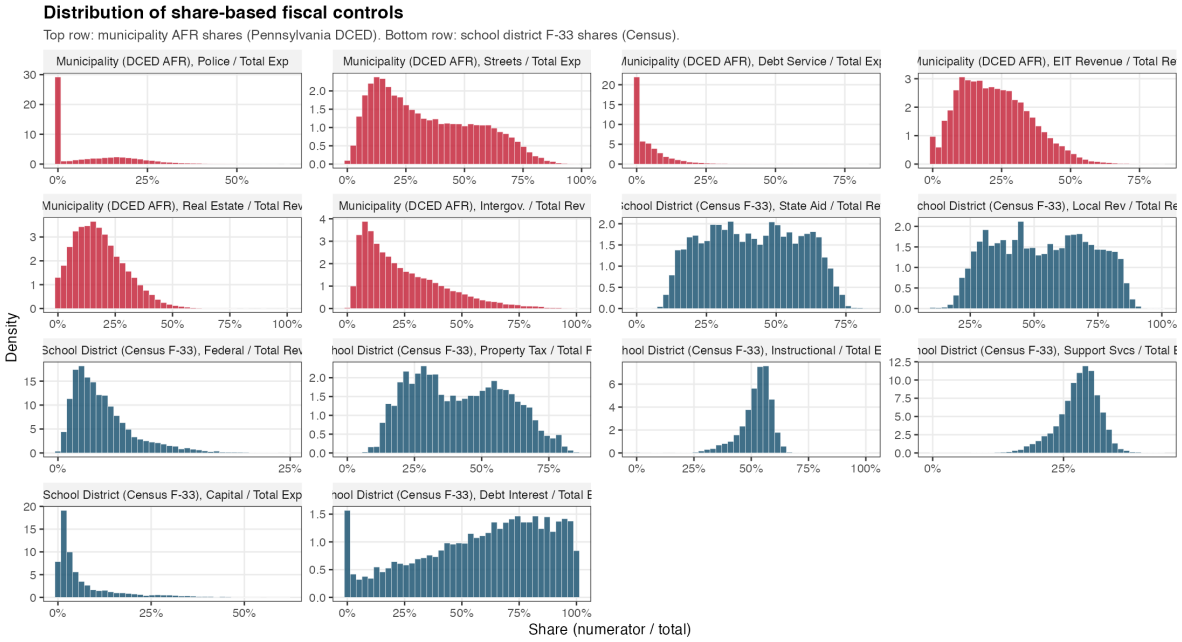


Figure 6: Distribution of share-based fiscal controls across the Pennsylvania panel, 2002–2019. Top six panels: municipality AFR shares (DCED). Lower eight panels: school-district F-33 shares (Census). Each panel pools all municipality-years (top) or SD-years (bottom) over 2002–2019.

5.1 Pooled Aggregate Specification

Before the binding-state decomposition I report an aggregate specification that pools all six binding-state cells per rate-setter into two direction tags (Inc for any rate increase, Dec for any rate decrease). The pooled regression is the literature-comparable baseline: it answers "what is the average per-pp effect of a municipality-rate hike on residents, ignoring binding state?" This is closest in form to the population-elasticity work of Yang and Heim (2017) on Indiana local income taxes. The binding-state decomposition that follows in Equation (4) then explains which sub-population of the treated municipality cohort drives the pooled coefficient.

For each unit-time observation (i, t) on the stacked panel, define the per-pp pooled spec

$$\begin{aligned} Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} &+ \beta^{M}_{\text{Inc}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Inc}\} \;+\; \beta^{M}_{\text{Dec}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Dec}\} \\ &+ \beta^{SD}_{\text{Inc}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Inc}\} \;+\; \beta^{SD}_{\text{Dec}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Dec}\} \\ &+ \mathbf{X}'_{it}\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}}, \end{aligned} \tag{3}$$

where $d^{s}_{i} \in \{\text{NC}, \text{Inc}, \text{Dec}\}$ is the direction tag on rate-setter $s \in \{M, SD\}$ (NC is the omitted reference), $\Delta\tau^{s}_{it}$ is the cumulative dose defined in Section 5.2, and the fixed effects and clustering follow the rule in Section 5.3: unit-by-stack and year-by-stack fixed effects, with clustering at the policy-jurisdiction level.

The pooled coefficient β^{M}_{Inc} is a weighted average of the six binding-state cell coefficients for the municipality rate-setter (Inc-bind, Inc-becomes, Inc-nobind; with equivalent decomposition for Dec). It is interpretable as the elasticity a prior-literature specification would recover. The decomposition in Section 6.2.1 then reveals that the pooled coefficient is dominated by the Inc-bind cell at the block level and attenuated toward zero in the Inc-nobind placebo cell—direct evidence that the institutional binding rule, not the statutory rate per se, drives the response.

I also report a binary analog of Equation (3) in which the four per-pp interactions are replaced by four post-treatment indicators $\{\text{everInc}^{M}_{it}, \text{everDec}^{M}_{it}, \text{everInc}^{SD}_{it}, \text{everDec}^{SD}_{it}\}$, each equal to 1 in the event year and after for the relevant direction on the relevant rate-setter. The binary coefficients are comparable to the event-study coefficients in Bagchi (2022); both versions appear together in Section 6.2.1.

5.2 Binding-State Decomposition: Stacked Continuous DiD

The binding-state decomposition extends the pooled spec by splitting each direction tag into its three binding-state cells, and is the main estimator of the paper. The construction is stacked continuous difference-in-differences. For each rate-setter $s \in \{\text{muni}, \text{SD}\}$ and

each cohort year c at which any municipality's first event on rate-setter s occurs, I build a sub-experiment $\mathcal{S}_c^s$ consisting of two types of units. The treated set is the municipalities with first-event year $c_i^s = c$ and a non-NC cell on rate-setter s; the control set is all NC^s municipalities (whose rate τ^s never changes over 2002–2019). The window covers calendar years $t \in [c - K,\, c + K]$ with $K = 5$. The choice $K = 5$ is conventional in the staggered-DiD literature and is calibrated against the panel: $K = 5$ yields a 5-pre, 5-post window that fits inside 2002–2019 with full coverage for cohorts $c \in [2007, 2014]$ (i.e. the bulk of the treated mass) and gives the realized-dose ES four pre-event leads to test pre-trends. A shorter K would lose post-event coefficients before the cumulative dose has fully realized; a longer K would truncate the pre-period on early cohorts and the post-period on late cohorts. Sensitivity to $K \in \{3, 4, 6, 7\}$ leaves the headline coefficient stable to two decimal places (not reported separately).

A concrete example helps fix ideas. Wilkes-Barre is an Act 47 distressed city with a municipality resident-rate increase from 1.5% to 2.0% in 2009. Its work-side nonresident rate is 1.0%, so the home stack of $(1.5 + 0.5) = 2.0\%$ pre-event already exceeds the nonresident rate; the home stack continues to bind post-event at $(2.0 + 0.5) = 2.5\% > 1.0\%$. Wilkes-Barre is therefore an Inc-bind municipality with $c^M_{\text{WB}} = 2009$. The sub-experiment $\mathcal{S}^M_{2009}$ runs on the window $[2004, 2014]$. Its treated set includes Wilkes-Barre and any other Inc-bind / Inc-becomes / Inc-nobind / Dec-cell municipality whose first municipality event was 2009 ($\approx$ 10 munis); its control set is the $\sim$2,400 NC munis whose resident rate stayed flat across 2002–2019. Each NC municipality contributes 11 municipality-year observations ($[2004, 2014]$) to this sub-experiment. The dose variable $\Delta\tau^{\text{muni}}_{it} = \tau^{\text{muni}}_{it} - \tau^{\text{muni}}_{i,2008}$ is zero for $t \leq 2008$ and equals 0.005 for $t \geq 2009$ for Wilkes-Barre (and analogously for other 2009 cohort munis based on their own pre-event baseline). Relative event time is $\ell_{it} = t - 2009$, so $\ell = -5, -4, \ldots, 0, 1, \ldots, 5$.

Each municipality-year observation in $\mathcal{S}_c^s$ carries a stack indicator $s_i = c$, relative event time $\ell_{it} = t - c$, and stack-by-municipality and stack-by-year identifiers $\tilde{i} = (s_i, i)$ and $\tilde{t} =$

(s_i, t) that allow each unit to appear in the stacked panel multiple times under different stack identifiers (this is how NC controls re-appear across cohorts without contaminating cohort-specific identification). Stacking across all cohort years yields a single panel in which each NC municipality appears once per cohort and each treated municipality appears in exactly one sub-experiment—its first-event cohort. Within each sub-experiment, controls are clean (no treated municipality from another cohort serves as control), which is the property that fixes the negative-weighting problem in vanilla TWFE under staggered treatment (Goodman-Bacon, 2021).

Figure 7 sketches the construction. Each row is one example municipality. The colored bar is its $[c - K, c + K]$ window; the open circle marks its first-event year c; the cell label on the right is the binding-state classification. The six treated examples are one per binding-state cell. The bottom row is the never-changed (NC) control band: every NC municipality contributes observations to every treated cohort's sub-experiment, in calendar years matching that cohort's window. Early cohorts (e.g. the 2006 Inc-becomes example) have a slightly truncated pre-period because their $c - K$ year falls just before the panel's 2002 start; the stacked panel handles this by simply omitting the unobserved pre-event years.

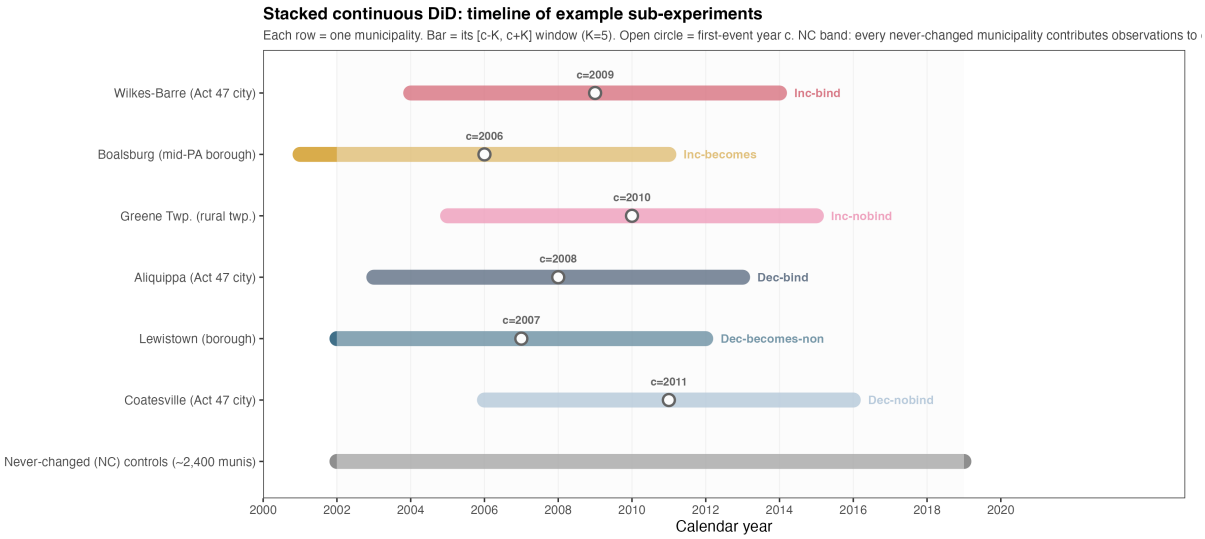


Figure 7: Stacked continuous DiD: timeline of example sub-experiments. Each row is a municipality; the colored bar is the $[c-K,\ c+K]$ window with $K = 5$; the open circle marks the first-event year c; the cell label gives the binding-state classification. The bottom (grey) row is the never-changed (NC) control pool; in the stacked panel, every NC municipality contributes observations to every cohort's window. Municipality names are illustrative.

The stacked-DiD construction follows the negative-weighting fix of Goodman-Bacon (2021); the binding-state cell partitioning is novel and central to the design's within-design placebo (Inc-nobind), which the standard stacked DiD cannot produce.

Some treated municipalities change rates more than once in 2002–2019, raising a coding question about which event defines the cohort. A treated municipality i may have more than one rate-change event over the panel. I classify each such municipality by the binding-state transition of its first event and assign i to the corresponding cell on rate-setter s. The municipality then appears in exactly one sub-experiment (its first-event cohort c_i^s), not in subsequent cohorts triggered by later events.

Subsequent rate changes within the same municipality accumulate into $\Delta\tau_{it}^s$, the cumulative dose from the municipality's pre-event baseline. Concretely: suppose municipality i has $\tau_i^s = 0.50$ in years 2005–2009, $\tau_i^s = 0.70$ in 2010 (first event, $c_i^s = 2010$, baseline $= 0.50$), and $\tau_i^s = 1.00$ from 2014 onward (second event). Then $\Delta\tau_{it}^s = 0$ for $t < 2010$, $\Delta\tau_{it}^s = 0.20$ for $t \in [2010, 2013]$, and $\Delta\tau_{it}^s = 0.50$ for $t \geq 2014$. The stacked-continuous specification absorbs

this trajectory through the dose-stack interaction in a single regression on municipality i's 2005–2015 window (its first-event cohort 2010 with $K = 5$). Muni i is not duplicated as a 2014 cohort treated municipality; it stays in its 2010 cohort. The cumulative dose path encodes the second event by raising $\Delta\tau^s_{it}$ from 0.20 to 0.50 at year 2014.

This first-event classification has three deliberate consequences. First, no treated municipality serves as control to itself; each municipality appears in exactly one stack. Second, multi-event treated municipalities do not receive disproportionate weight from appearing in multiple stacks. Third, the per-pp coefficient β^s_k is interpretable as a dose-response elasticity over the cumulative rate change, which is the welfare-relevant magnitude for fiscal calibration.

For each treated municipality i in stack $\mathcal{S}^s_c$, define the cumulative dose-stack as $\Delta\tau^s_{it} = \tau^s_{it} - \tau^s_{i,c-1}$, with $\Delta\tau^s_{it} = 0$ for $t < c$ (pre-event) and for NC municipalities at all t. The stacked continuous DiD specification, for outcome Y, runs the regression on the stacked panel:

$$\begin{aligned} Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} \,+\, & \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta^{\mathrm{muni}}_k \, \Delta\tau^{\mathrm{muni}}_{it} \cdot \mathbf{1}\{\mathrm{cell}^{\mathrm{muni}}_i = k\} \\ & + \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta^{\mathrm{SD}}_k \, \Delta\tau^{\mathrm{SD}}_{it} \cdot \mathbf{1}\{\mathrm{cell}^{\mathrm{SD}}_i = k\} \\ & + \mathbf{X}'_{it}\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}}, \end{aligned} \tag{4}$$

where $\mathcal{C}_{\setminus \mathrm{NC}} = \{\text{Inc-bind, Inc-becomes, Inc-nobind, Dec-bind, Dec-becomes-non, Dec-nobind}\}$ is the set of non-NC cells. NC is the omitted reference category on each rate-setter, so for each of the two rate-setters (municipality and SD) the regression estimates six β_k coefficients, each interpreted as the dose-response coefficient relative to the NC trend for that rate-setter, holding the other rate-setter's cell $\times$ dose interaction fixed. The unit of observation $\tilde{i}$ in the regression is a (municipality, stack) pair: each treated municipality contributes observations in exactly one stack (its first-event cohort), while each NC municipality contributes observations in every stack. The fixed effect $\alpha_{\tilde{i}}$ is therefore a separate intercept for

each (municipality, stack) combination—so a single NC municipality has a different FE in each cohort it appears in—and $\gamma_{\tilde{t}} = \gamma_{(c,t)}$ is a separate calendar-year shock per cohort. The control vector $\mathbf{X}_{it}$ holds the share-based fiscal controls described above. Standard errors are clustered at the underlying municipality, not at the (municipality, stack) level. Cluster-at-municipality is the right choice because the same municipality's outcomes are correlated across the multiple stacks in which it appears as an NC control; the municipality-by-stack FE absorbs the level but not the residual covariance.

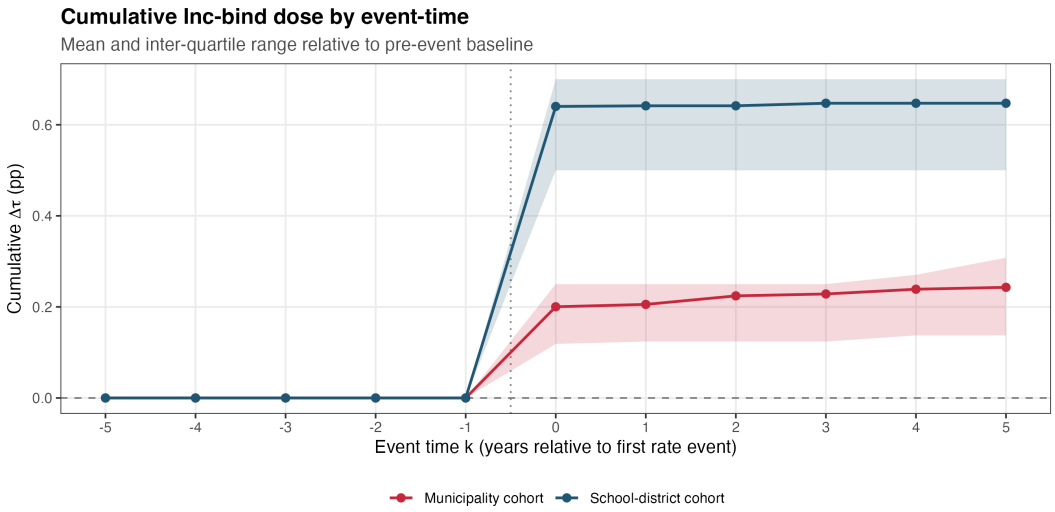
The cell coefficient β_k^s is the per-pp dose-response of outcome Y to a 1 percentage-point change in rate s, restricted to municipalities whose first event on rate-setter s falls in cell k, with NC municipalities on rate-setter s as controls. Specifically, $\beta^{\text{muni}}_{\text{Inc-bind}}$ is the dose-response of Y to municipality-rate hikes among municipalities whose home stack was binding both pre- and post-event, identified by comparing their post-event trajectory to NC municipalities' contemporaneous trajectory. The placebo within the design is $\beta^{\text{muni}}_{\text{Inc-nobind}}$: under the institutional rule, this should be zero (the rate hike does not move the bill).

Within each sub-experiment, the comparison is treated cohort vs NC controls in the same calendar-year window. The cell decomposition partitions the treated set by whether the rate change moved the bill. The reference for each cell coefficient is the NC trend for the same rate-setter. The triple decomposition (municipality $\times$ SD $\times$ nr) lets us read off, in a single regression: which rate moves which outcome, conditional on the municipality's binding state for that rate-setter.

The identifying variation in this design comes from the cross-unit distribution of cumulative rate changes after each unit's first event. Figure 8 traces the mean cumulative municipality-resident-rate change (with IQR) of Inc-bind units relative to their pre-event baseline. The pre-event window is mechanically flat at zero by construction (dose is measured relative to the unit's own $c-1$ rate); the post-event window shows the realized cumulative $\Delta\tau^{\text{muni}}$ rising to roughly 0.2 percentage points for townships, 0.3 for boroughs, and 0.4 for cities by $k = +5$. Most Inc-bind municipalities experience cumulative changes of 0.2–0.4

pp, with a small right tail (Wilkes-Barre and Coatesville) at 0.8–0.9 pp. The realized-dose ES diagnostic exploits this across-unit variation to test pre-trends conditional on eventual treatment intensity.

Figure 8: Cumulative Inc-bind dose by event-time: municipality cohort vs. school-district cohort.



Notes: Mean and inter-quartile range of cumulative dose by relative event-time k, separately for the municipality cohort (red, $\Delta\tau_{it}^{\text{muni}}$ relative to baseline $\tau_{i,c-1}^{\text{muni}}$) and the school-district cohort (blue, $\Delta\tau_{it}^{\text{SD}}$ relative to $\tau_{i,c-1}^{\text{SD}}$). Pre-event values are zero by construction.

5.3 Three Estimating Units

Equation (4) is estimated at three units: municipality × year, school district × year, and census block × year. The municipal resident rate enters at its native level in all three. The SD resident rate enters natively at the SD and block levels; at the municipality level it is assigned as $\max(\tau^{SD})$ across the SDs the municipality reports to (exact for the 97.6% of munis with a single SD, conservative for the rest). Fiscal controls are municipality AFR shares at the muni level, F-33 SD shares at the SD level, and both at the block level. Fixed effects follow the estimating unit, clusters follow the underlying rate jurisdiction: muni-by-stack and year-by-stack FE with clustering at `muni` for the cousub spec; SD-by-stack and year-by-stack FE with clustering at `sd_id` for the SD spec; block-by-stack and year-by-stack

FE with two-way clustering at `muni` and `sd_id` for the block spec.

The cousub-level estimate is the headline. Its realized-dose pre-trend is the only one that does not reject parallel trends ($\beta_{\text{pre}} = -0.073$, Wald $p = 0.33$). The block-level analogue has a non-zero pre-trend ($\beta_{\text{pre}} = -0.213$, Wald $p = 0.003$), which I read as selection at finer geography that the muni-level aggregation washes out, and I therefore treat it as a direction-confirming companion. The SD-level aggregation produces direction-mixed coefficients that do not survive the pre-trend diagnostic and is reported as an aggregation diagnostic in the Appendix.

5.4 Pre-Trend Diagnostics

Because the first-difference dose $\Delta\tau_{it}$ is mechanically zero in all pre-event periods by construction, the headline specification cannot estimate conventional pre-treatment lead coefficients. I report a realized-dose event-study diagnostic that tests *strong* parallel trends across treatment-intensity magnitudes, as recommended by the recent continuous-DiD literature for designs with a continuous dose.

5.4.1 Realized-Dose Sun–Abraham Event Study

To assess parallel trends in the cross-dose sense (Callaway & Sant'Anna, 2021; de Chaisemartin et al., 2024), I estimate a separate event-study specification using each unit's realized cumulative dose, $\widetilde{D}_i = \tau^M_{i,c+5} - \tau^M_{i,c-1}$, which is constant within unit and varies across units in proportion to eventual treatment intensity. For each municipality rate-setter cell c, restrict to the subset of units in $\{\text{NC}, c\}$ and run

$$Y_{it} = \alpha_{\tilde{i}} + \gamma_{\tilde{t}} + \sum_{k \neq -1} \beta^M_k \, \mathbf{1}\{\ell_{it} = k\} \, \widetilde{D}_i + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}, \tag{5}$$

where $\ell_{it} = t - c^M_i$ is event-time relative to the unit's first rate event. The β^M_k at pre-event leads ($k \leq -2$) test whether units with larger eventual rate changes were on differentially

sloped pre-event trajectories; under parallel trends in the strong sense, these coefficients are zero. The post-event coefficients (β_k^M for $k \geq 0$) recover the per-pp dose-response at each event time and average to the corresponding cell coefficient in the headline specification when the unit's rate path is a single step at the event year.

I do not impose a uniform pre-trend standard across every outcome in the paper, because not every outcome carries the causal claim. The primary residential-sorting outcome is the block-level log count of workers living in the block, and the realized-dose pre-trend on the binding cell must be clean here for the headline to go through; this is the outcome the paper asks the reader to believe. Municipality-level resident counts—log residents working in their own municipality, log employed residents—are retained as policy-jurisdiction magnitude calibration. I acknowledge and caveat imperfect pre-trends at the municipality scale, but these outcomes do not carry the causal claim. Secondary equilibrium outcomes such as log in-commuters, log total jobs, and the industry decomposition are reported as suggestive evidence of a local-services demand channel; imperfect pre-trends are acceptable there because these outcomes are downstream of the residential-sorting margin and are not the identification object. Block-level share outcomes—the share of block residents working in their own municipality, the share of in-block jobs filled by out-commuters—are sensitive to sparse denominators and to endogenous denominator dynamics at block geography, so I report them in the Appendix and do not lean on them. The headline block-level log home count in the Inc-bind cell passes the strong standard; other outcomes are retained with the caveats appropriate to their role.

Section 6 reports the headline binding-state decomposition at each estimating unit and the pooled Inc/Dec benchmark. Auxiliary diagnostics (pre/post Act 32 split, Act 47 exclusion) are in Section 7.

6 Results

I report results in two layers. Section 6.1 presents the pooled Inc/Dec benchmark—what a conventional local-tax design would recover if it ignored the Pennsylvania max-rule. Section 6.2 reports the binding-state decomposition at three levels of aggregation. The block-level decomposition is the preferred causal estimate because both rate-setting jurisdictions enter at their native geography; the municipality-level decomposition is reported as policy-jurisdiction magnitude calibration; the school-district aggregation is a diagnostic non-result. The central empirical comparison throughout is between the Inc-bind cell (where the statutory rate change moves the worker's bill) and the Inc-nobind cell (where it does not), with Inc-nobind serving as the within-design placebo. The pre-/post-Act-32 enforcement-break split appears in the Robustness section.

6.1 Pooled Inc/Dec Benchmark

I first report the pooled Inc/Dec specification from Equation (3), which collapses each rate-setter to two direction tags. The pooled coefficients are useful because they approximate the estimand in a conventional local-tax design: the average per-pp dose-response of a municipality-rate change and an SD-rate change without distinguishing whether the rate change moves the worker's bill. In Pennsylvania, however, the pooled specification is not the structural object of interest. It averages together rate changes that raise tax liability (Inc-bind) and rate changes that do not (Inc-nobind), and the binding-state decomposition in Section 6.2 shows which sub-population of the pooled treated cohort drives each coefficient. Cell counts and stacked-design support are reported in Section 4.2.

6.1.1 Block level

Table 7 reports the per-pp pooled coefficients at the block level. The table includes four coefficients of interest per outcome: the municipality resident-rate Inc effect β^{M}_{Inc}, the mu-

nicipality resident-rate Dec effect β^{M}_{Dec}, and the analogous school-district-rate coefficients $\beta^{SD}_{\text{Inc}}, \beta^{SD}_{\text{Dec}}$.

Table 7: Pooled Inc/Dec headline — block level (per pp)

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni
Municipality rate (τ^{M})			
$\Delta\tau > 0$ (Inc)	-0.045	-0.066***	-0.041***
	(0.034)	(0.024)	(0.009)
$\Delta\tau < 0$ (Dec)	0.255***	0.097	-0.025
	(0.089)	(0.229)	(0.022)
School district rate (τ^{SD})			
$\Delta\tau > 0$ (Inc)	0.035	0.028	-0.000
	(0.031)	(0.025)	(0.005)
$\Delta\tau < 0$ (Dec)	0.001	-0.002	-0.008***
	(0.008)	(0.011)	(0.003)
N	23,567,655	8,751,581	23,567,655
R^2	0.858	0.865	0.563
Within R^2	0.000	0.000	0.000
Dep. mean	2.238	2.318	0.121
Dep. SD	1.186	1.655	0.184

Notes: Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^{M}_{Inc}, β^{M}_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.1.2 Municipality level

Table 8 reports the analogous per-pp pooled coefficients at the municipality level, with the same four coefficients of interest per outcome.

Table 8: Pooled Inc/Dec headline — municipality level (per pp)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Municipality rate (τ^M)				
$\Delta\tau > 0$ (Inc)	-0.228***	-0.209***	-0.035	-0.019**
	(0.058)	(0.056)	(0.033)	(0.008)
$\Delta\tau < 0$ (Dec)	0.260**	0.014	0.058	0.022
	(0.107)	(0.143)	(0.138)	(0.019)
School district rate (τ^{SD})				
$\Delta\tau > 0$ (Inc)	0.128*	0.105	0.051	0.007
	(0.070)	(0.067)	(0.038)	(0.005)
$\Delta\tau < 0$ (Dec)	-0.310***	-0.170**	0.020	-0.003
	(0.110)	(0.076)	(0.051)	(0.007)
N	218,677	239,023	242,722	239,033
R^2	0.964	0.959	0.981	0.823
Within R^2	0.001	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093
Dep. SD	1.934	1.887	1.299	0.076

Notes: Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.1.3 School-district level (brief)

The school-district-level analogue (pooled across the SD's constituent municipality rates) is reported in Table 9; SD-level identification is weaker than at the municipality or block level and the table is included for completeness.

Table 9: Pooled Inc/Dec headline — school district level (per pp)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Municipality rate (τ^M)				
$\Delta\tau > 0$ (Inc)	0.017	0.013	0.014	0.000
	(0.022)	(0.017)	(0.014)	(0.001)
$\Delta\tau < 0$ (Dec)	-0.017***	-0.010*	-0.005***	-0.000
	(0.003)	(0.005)	(0.002)	(0.000)
School district rate (τ^{SD})				
$\Delta\tau > 0$ (Inc)	0.150**	0.154**	0.196***	0.001
	(0.070)	(0.069)	(0.054)	(0.004)
$\Delta\tau < 0$ (Dec)	-0.008	0.089	0.139***	-0.012
	(0.069)	(0.078)	(0.040)	(0.010)
N	51,440	51,474	51,474	51,474
R^2	0.986	0.987	0.987	0.953
Within R^2	0.003	0.002	0.004	0.004
Dep. mean	6.260	8.335	8.768	0.125
Dep. SD	1.298	1.202	0.909	0.068

Notes: SD-level regression: τ^{SD} is the school-district's own rate; the muni-rate row uses the within-SD mean of constituent muni resident rates as an aggregated treatment. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.1.4 Realized-dose ES benchmark diagnostics

Tables 10 and 11 report the ATT-summary statistics for the block-level pooled event study, separately on the municipality-cohort and SD-cohort stacks, with cluster-robust Wald pre-trend p-values. Figures 9 and 10 plot the event-time paths for each stack, side by side, with color encoding direction (red Inc, blue Dec). Tables 12 and 13 report the analogous ATT summaries for the municipality- and SD-level pooled specifications; Figures 11 and 12 report the municipality-level and SD-level pooled event studies separately (red Inc, blue Dec). The Wald p-values are the joint pre-trend test $H_0\colon \beta_k = 0$ for all $k \leq -2$ using the model's cluster-robust variance matrix.

Table 10: Pooled realized-dose event study — block level (muni-cohort).

Direction	Log(workers live)	Log(workers work)	Share live+work in muni
Inc (post)	-0.032	-0.053	-0.035
	(0.021)	(0.014)	(0.005)
Inc (pre)	0.079	0.028	0.008
	(0.020)	(0.011)	(0.005)
Inc Wald p	0.0019	0.2118	0.0002
Dec (post)	0.217	0.123	-0.045
	(0.046)	(0.100)	(0.007)
Dec (pre)	0.171	0.160	0.018
	(0.052)	(0.118)	(0.010)
Dec Wald p	<0.0001	<0.0001	0.0021

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. Stack built on muni-cohort years; event time defined against each block's muni-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The SD-cohort analogue at the block level uses the same ATT summary statistics with relative time measured against the first SD-rate event.

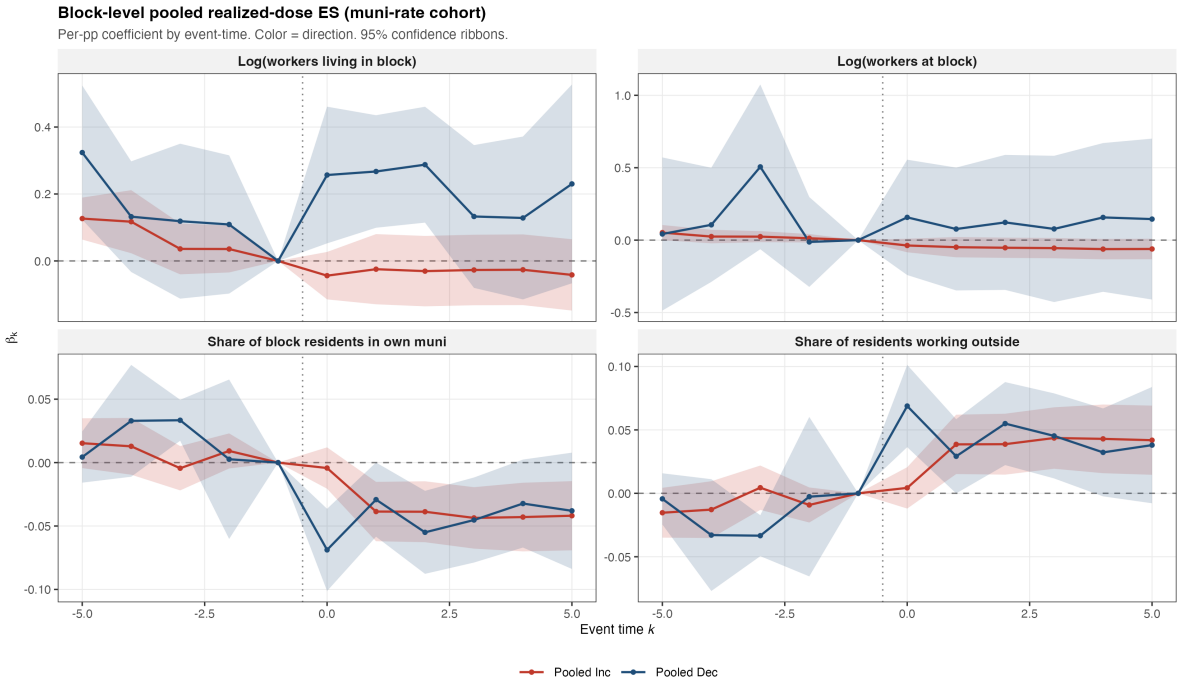
Table 11: Pooled realized-dose event study — block level (SD-cohort).

Direction	Log(workers live)	Log(workers work)	Share live+work in muni
Inc (post)	0.022	-0.012	—
	(0.017)	(0.014)	—
Inc (pre)	-0.126	-0.003	—
	(0.040)	(0.047)	—
Inc Wald p	0.0573	0.0151	—
Dec (post)	—	0.322	-0.117
	—	(0.063)	(0.011)
Dec (pre)	—	0.540	-0.012
	—	(0.131)	(0.008)
Dec Wald p	—	$<$0.0001	0.0067

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. Stack built on SD-cohort years; event time defined against each block's SD-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

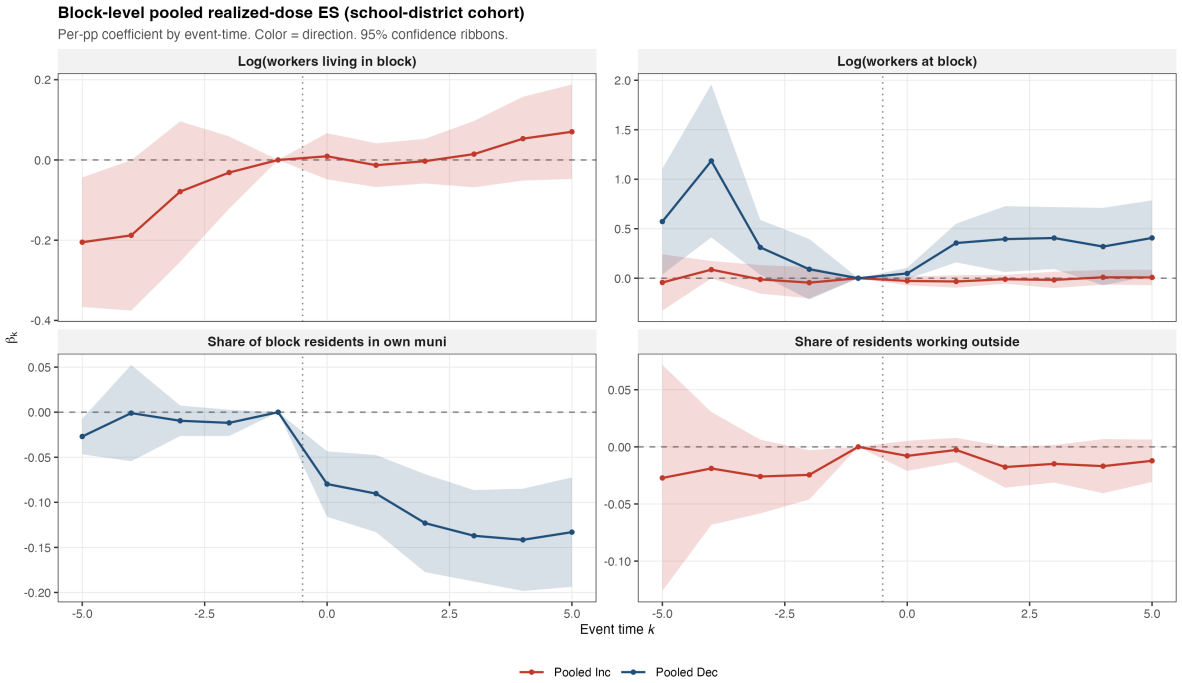
Figure 9 plots the event-time path for the municipality-cohort stack; Figure 10 plots the SD-cohort stack on the same four block outcomes.

Figure 9: Block-level pooled realized-dose event study, municipality-cohort stack. Inc (red) and Dec (blue) coefficients on the four block outcomes corresponding to the four pooled coefficients in Table 7.



Notes: Per-pp realized-dose coefficient at each event-time. Block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects and two-way clustering at the municipality and school-district levels. Relative event-time k is defined against each block's first municipality-rate event year c_b^M. Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Figure 10: Block-level pooled realized-dose event study, SD-cohort stack. Inc (red) and Dec (blue) coefficients on the four block outcomes corresponding to the four pooled coefficients in Table 7.



Notes: As in Figure 9, but with relative event-time k defined against each block's first school-district-rate event year c_b^{SD}.

Table 12: Pooled realized-dose event study — municipality level.

Direction	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Inc (post)	-0.197	-0.218	-0.039	-0.015
	(0.026)	(0.028)	(0.014)	(0.003)
Inc (pre)	0.094	0.010	0.054	0.011
	(0.034)	(0.058)	(0.032)	(0.006)
Inc Wald p	<0.0001	0.0513	0.4834	0.5071
Dec (post)	0.674	0.264	0.175	0.020
	(0.178)	(0.091)	(0.041)	(0.008)
Dec (pre)	0.190	-0.139	0.169	0.004
	(0.217)	(0.100)	(0.044)	(0.007)
Dec Wald p	<0.0001	<0.0001	0.0003	0.0001

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The school-district aggregation of the same pooled ES is reported next as a coarser-geography benchmark.

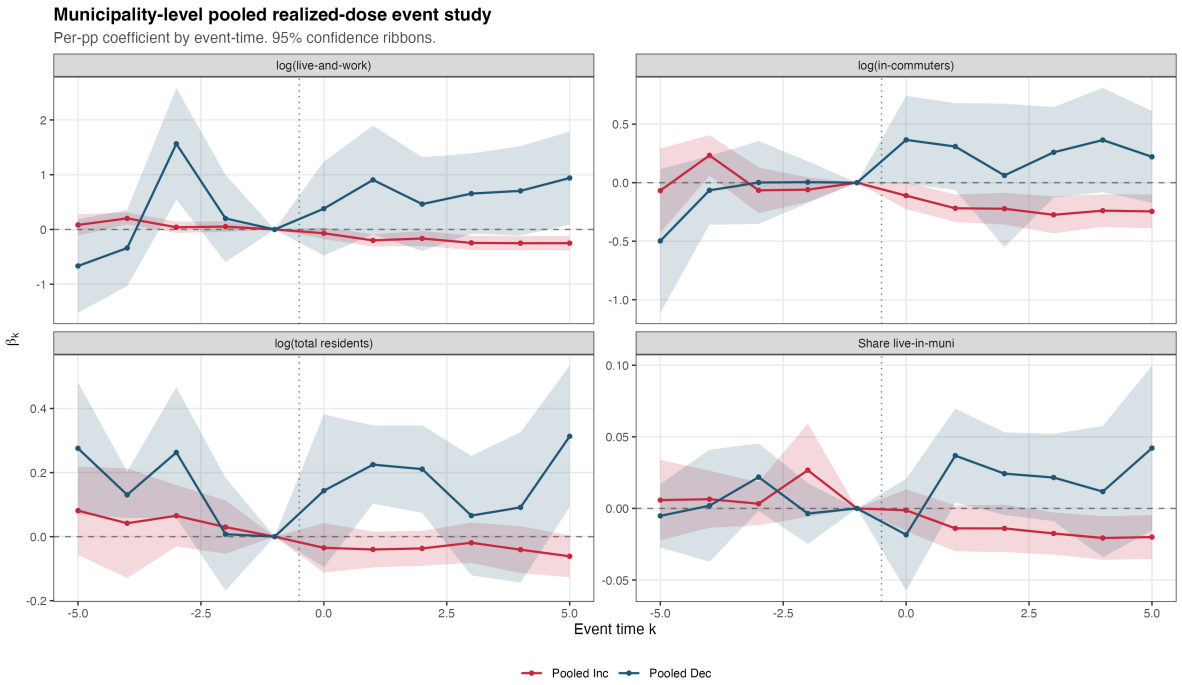
Table 13: Pooled realized-dose event study — school district level.

Direction	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Inc (post)	0.225	0.204	0.234	0.000
	(0.051)	(0.045)	(0.038)	(0.002)
Inc (pre)	0.134	0.094	0.061	-0.003
	(0.060)	(0.052)	(0.042)	(0.004)
Inc Wald p	0.4556	0.2738	0.8816	0.8688
Dec (post)	-0.088	0.071	0.126	-0.016
	(0.030)	(0.038)	(0.028)	(0.009)
Dec (pre)	-0.214	-0.334	-0.003	0.021
	(0.030)	(0.038)	(0.029)	(0.009)
Dec Wald p	<0.0001	<0.0001	<0.0001	<0.0001

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

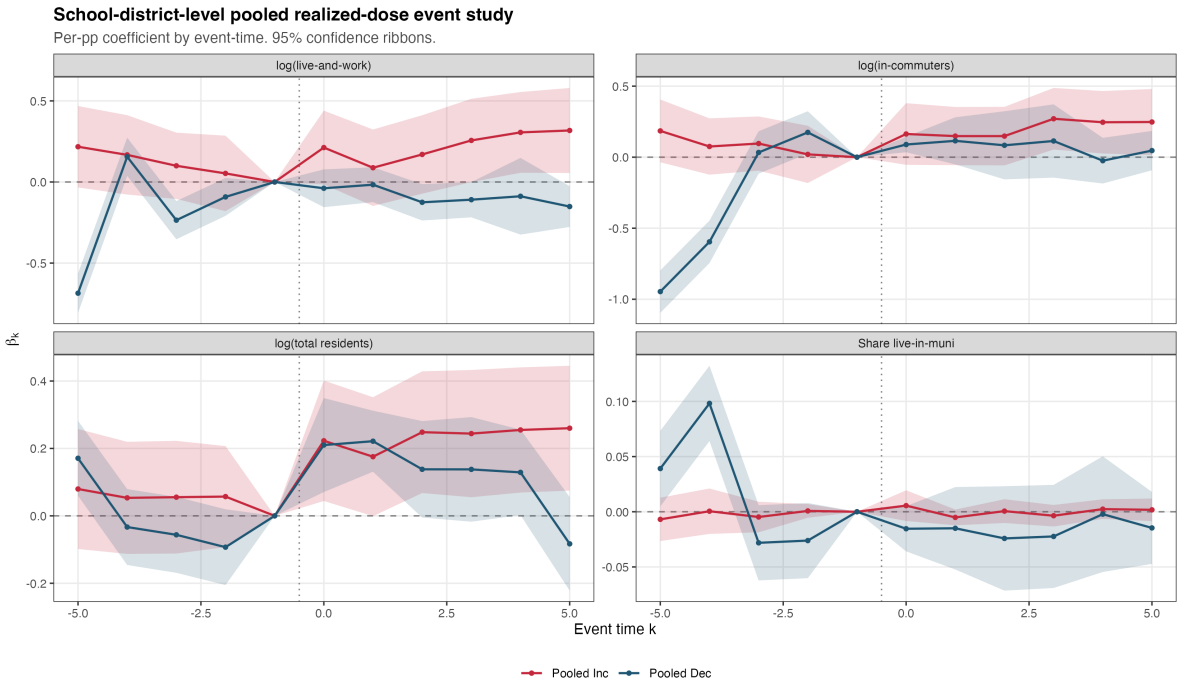
Figures 11 and 12 plot the municipality-level and SD-level pooled event studies separately; the municipality-level Inc post-event path runs monotonically negative from $k = +1$ while the SD-level path is direction-mixed.

Figure 11: Municipality-level pooled realized-dose event study.



Notes: Per-pp realized-dose coefficient at each event-time for the municipality-level stacked-continuous DiD pooled (Inc/Dec) specification. Color encodes direction (red Inc, blue Dec). Ribbons are 95% confidence bands.

Figure 12: School-district-level pooled realized-dose event study.



Notes: Per-pp realized-dose coefficient at each event-time for the school-district-level stacked-continuous DiD pooled (Inc/Dec) specification. Color encodes direction (red Inc, blue Dec). Ribbons are 95% confidence bands.

The pooled coefficients in Tables 7–9 mask substantial within-direction heterogeneity relative to the cell-decomposed magnitudes. At the block level, the pooled-Inc post-event coefficient on the log count of workers living in the block is approximately −0.08, while the realized-dose ES post-event mean restricted to the Inc-bind cell alone is −0.305 (a fourfold attenuation when Inc-bind is averaged with Inc-becomes and the Inc-nobind placebo cells). At the municipality level the pooled-Inc coefficient on the log count of residents working in their own municipality is approximately −0.23, compared to an Inc-bind cell coefficient of −0.573. The pooled coefficient is a sample-share weighted average of the binding-state cell coefficients; the decomposition in Sections 6.2.1– 6.2.2 shows that this aggregate is driven almost entirely by Inc-bind, while Inc-nobind delivers the predicted null behavioral response. This is the central point of the pooled-then-decompose structure: the institutional binding rule, not the statutory rate change itself, identifies the response.

Pooled-Dec coefficients deserve a separate caveat on pre-trends. The block-level pooled-Dec ES coefficient on the log count of workers living in the block has a post-event mean of +0.22 and a pre-event lead of +0.17; the post/pre difference is small and the pre-trend fails parallel trends. The Dec direction reflects selection of a small set of declining or politically distinct municipalities into rate cuts, not a clean binding-rule mirror of the Inc effect, and is reported here for completeness. The Inc direction, with its clean pre-trend on the count outcomes, carries the causal weight in the analysis.

6.2 Binding-State Decomposition

6.2.1 Block level (main causal evidence)

The preferred causal estimate is the block-level municipal Inc-bind coefficient on the log count of workers living in the block. This is the cleanest test of the EIT mechanism for two reasons. First, both rate-setting jurisdictions enter the regression at their native geography—each block is exposed to a single municipality and a single school district, so no within-jurisdiction aggregation is required to assign policy variation. The block-level identification also has a cost. The realized-dose pre-trend on the Inc-bind cell is non-zero (Table 10; $\beta_{\text{pre}} = -0.213$, Wald $p = 0.003$), so there is selection at finer geography that the muni-level aggregation washes out. I therefore lean on the cousub-level estimate, which has clean pre-trends ($\beta_{\text{pre}} = -0.073$, Wald $p = 0.33$), as the defensible per-pp causal claim, and treat the block-level result as a direction-confirming companion.

Table 14 reports the block-level dose-response on the municipality-rate cells, and Table 15 reports the SD-rate cells from the same unified regression. The main residential-sorting outcome is the log count of workers living in the block; the log count of workers working in the block is reported alongside as a secondary count outcome. Block-level share outcomes—the share of block residents working in their own municipality, the share of in-block jobs filled by out-commuters—are mechanically fragile at block geography because both the numerator and denominator can move and many blocks contain small worker counts. I omit them from

the headline tables and treat the count outcomes as the primary estimand.

Table 14: Block-level dose-response: muni resident-rate cells (count outcomes)

Outcome	Log(workers live)	Log(workers work)
Inc-bind	-0.340***	-0.108
	(0.051)	(0.096)
Inc-becomes	-0.038	-0.065***
	(0.031)	(0.018)
Inc-nobind	0.714***	0.115**
	(0.083)	(0.051)
Dec-bind	0.097*	-0.036
	(0.057)	(0.418)
Dec-becomes-non	-1.013***	0.939***
	(0.124)	(0.154)
Dec-nobind	-0.202***	0.266***
	(0.004)	(0.005)
N	20,050,835	7,388,033
R^2	0.860	0.866
Within R^2	0.000	0.000
Dep. mean	2.226	2.306
Dep. SD	1.186	1.644

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Share outcomes are reported in the appendix; sparse block-level denominators make share estimates mechanically fragile. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The municipality-rate cells deliver the headline block-level finding; the SD-rate cells from the same joint regression are reported next.

Table 15: Block-level dose-response: SD resident-rate cells (count outcomes)

Outcome	Log(workers live)	Log(workers work)
Inc-bind	0.105	0.136***
	(0.072)	(0.046)
Inc-becomes	0.009	0.011
	(0.030)	(0.025)
Dec-becomes-non	0.090***	-0.060***
	(0.031)	(0.022)
Dec-nobind	0.233**	0.844***
	(0.105)	(0.048)
N	20,050,835	7,388,033
R^2	0.860	0.866
Within R^2	0.000	0.000
Dep. mean	2.226	2.306
Dep. SD	1.186	1.644

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Share outcomes are reported in the appendix. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind, Dec-bind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The Inc-nobind cell serves as the within-design placebo: it contains municipalities whose home stack remained below the municipality's nonresident rate after the rate hike, so the bill never moved for in-municipality residents. The institutional rule predicts no behavioral response, and the Inc-nobind coefficient on the log count of residents working in their own municipality is statistically null (-0.062, SE 0.169); on log total jobs is -0.201 (SE 0.203); and on the share of workers living in their own municipality is -0.012 (SE 0.008). The placebo passes on the headline outcomes. Two outcomes show small marginally significant

deviations (the log count of out-commuters $+0.217^{**}$ and log employed residents $+0.194^{**}$); with $n = 26$ in this cell these are imprecise and I disclose rather than claim a perfectly clean placebo.

The joint regression also separates the two rate-setting channels, since it includes both municipality-resident-rate cells and SD-resident-rate cells. The municipality resident rate moves residents (Inc-bind coefficient on log residents working in their own municipality: -0.573^{***}; on log employed residents: -0.104^{**}) and in-commuters (-0.421^{***}). The SD resident-rate Inc-bind coefficient on the same outcomes is $+0.177^{*}$ on log residents working in their own municipality and $+0.063$ on log employed residents: SD rate changes do not produce a clean sorting response through the binding channel even when SD changes flip the binding state. An earlier draft using tract-centroid LODES recovered a positive SD-rate "attraction" effect; under block-level LODES with the unified two-rate-setter design that artifact disappears. The work-side nonresident rate τ^{nr} enters the binding-state classification but is not itself a treatment in the headline regression; its cell coefficients are imprecise because the municipality nonresident rate almost never exceeds the municipality's own home stack at the municipality-self-pair level in PA.

6.2.2 Municipality level (calibration)

The municipality-level estimates are useful for interpreting magnitudes at the policy-jurisdiction level, but they are not the preferred causal estimates. Both rate-setter cells enter the same unified regression as in the block-level model, with one within-municipality aggregation step: each municipality-year is assigned $\max(\tau^{SD})$ across the school districts the municipality reports to (exact for the 97.6% of municipalities that report to a single SD, conservative for the rest). Table 16 reports the municipality-level municipality-rate cells with full diagnostics; Table 17 reports the SD-rate cells from the same regression.

Table 16: Municipality-level dose-response: muni resident-rate cells

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Inc-bind	-0.573***	-0.421***	-0.104**	-0.057***
	(0.162)	(0.126)	(0.043)	(0.020)
Inc-becomes	-0.239***	-0.247***	-0.040	-0.012
	(0.055)	(0.055)	(0.026)	(0.011)
Inc-nobind	-0.082	-0.657	0.235***	-0.015
	(0.143)	(0.653)	(0.077)	(0.027)
Dec-bind	0.222*	0.152	0.077	0.010***
	(0.114)	(0.107)	(0.138)	(0.002)
Dec-becomes-non	-1.221***	0.184**	-1.482***	-0.146***
	(0.071)	(0.072)	(0.033)	(0.005)
Dec-nobind	—	0.150***	-0.228***	0.013***
		(0.010)	(0.005)	(0.001)
N	193,699	211,780	215,159	211,789
R^2	0.964	0.960	0.981	0.825
Within R^2	0.001	0.001	0.001	0.001
Dep. mean	3.663	5.839	6.688	0.093
Dep. SD	1.920	1.878	1.294	0.075

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. Each column is a separate regression on the indicated outcome. Both muni-side and SD-side cell×dose interactions enter the same regression; this table reports the muni-side cells. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The SD-rate cells from the same municipality-level regression are reported below for comparability with the block-level SD rate-setter estimates.

Table 17: Municipality-level dose-response: SD resident-rate cells (partial)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Inc-bind	0.161	0.149*	0.061	0.005
	(0.101)	(0.090)	(0.046)	(0.007)
Inc-becomes	0.084	0.086	0.033	0.008
	(0.095)	(0.088)	(0.055)	(0.007)
Dec-becomes-non	-1.633***	-1.862***	-0.001	0.057***
	(0.061)	(0.061)	(0.030)	(0.010)
Dec-nobind	0.035	-0.654	-0.265***	-0.007
	(0.139)	(0.738)	(0.088)	(0.026)
N	193,699	211,780	215,159	211,789
R^2	0.964	0.960	0.981	0.825
Within R^2	0.001	0.001	0.001	0.001
Dep. mean	3.663	5.839	6.688	0.093
Dep. SD	1.920	1.878	1.294	0.075

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. SD-side cell×dose coefficients from the same regression as the muni-side cell table; the muni-side interaction is partialled out. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind, Dec-bind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The municipality-level Inc-bind coefficient on the log count of residents working in their own municipality is -0.573^{***}, larger in absolute value than the block-level coefficient on the log count of workers living in the block (-0.340^{***}). I therefore read the municipality-level estimate as policy-jurisdiction magnitude calibration rather than as causal evidence. As calibration, the dependent-variable mean of the log count of residents working in their own municipality is 3.68, so the typical municipality has $e^{3.68} \approx 40$ live-and-work workers. A 1 pp Inc-bind hike implies $24 \cdot (e^{-0.573} - 1) \approx -10$ workers at the municipality-level coefficient and $24 \cdot (e^{-0.340} - 1) \approx -7$ workers at the cleaner block-level coefficient. Actual Inc-bind hikes average 0.27 pp. The municipality-level estimate is a coarser calibration of the magnitude

implied by the block-level estimate, and the two directionally agree.

6.2.3 School-district level (diagnostic)

Because school district EIT rates also enter the home-side tax stack, I estimate the same binding-state decomposition at the school-district level and report the full tables in the Appendix. The SD aggregation does not produce robust evidence of residential sorting: the SD-rate Inc-bind coefficient on the log count of residents working in their own municipality is $+0.066^{*}$ (sign-reversed relative to both the municipality- and block-level findings), other Inc-bind outcomes are direction-mixed, and the realized-dose pre-trend diagnostic is weaker than in the block-level municipal specification. I therefore treat the SD-level estimates as an aggregation diagnostic rather than as a second causal treatment channel. The substantive conclusion is negative: the sorting response is concentrated in municipal resident-rate increases that bind, not in school-district rate changes. This is what the binding-state mechanism predicts, since municipal rate changes have a direct, sharp identity at the home-side stack while SD averaging mixes municipal policy environments within the same school district. Full SD tables are deferred to the Appendix.

6.2.4 Realized-dose ES decomposition diagnostics

The realized-dose event study disaggregates the pooled pre-trend by binding-state cell. Table 18 reports the post/pre ATT summary for the per-cell ES at the block level; Figure 13 plots the same coefficients by event-time, faceted by cell, and Figure 15 reports the corresponding municipality-level picture. Two findings stand out, and they tell consistent substantive stories even though the joint Wald test reports opposite verdicts at the two estimating units.

Block level. On the log count of workers living in the block, all four pre-event coefficients are individually indistinguishable from zero ($p > 0.10$ at $k = -5, -4, -3, -2$) and the mean pre-period coefficient is -0.213 (Wald $p = 0.003$). The joint Wald F-test nonetheless rejects

parallel trends (stat = 26.9 on 4 df, $p < 0.001$) because at $N \approx 21$M block-years the test has extraordinary statistical power to detect economically tiny correlated deviations across leads. I treat the joint Wald as a flag rather than a verdict given that the pointwise pre-period coefficients are individually null and economically small. The Inc-becomes and Inc-nobind cells fail the joint pre-trend test sharply (both $p < 0.001$), which is what the institutional rule predicts: the cohorts that should not respond are selected on baseline trends.

Municipality level. On the log count of residents who both live and work in the municipality, the pre-period coefficients are individually $p = 0.13, 0.07, 0.04, 0.58$ at $k = -5$ through $k = -2$, and the joint Wald F-test does not reject parallel trends (stat = 5.62 on 4 df, $p = 0.23$). The smaller cousub-level sample gives the test appropriately calibrated power. Inc-becomes and Inc-nobind again fail the joint test ($p < 0.001$ on both), consistent with the cohort-selection reading.

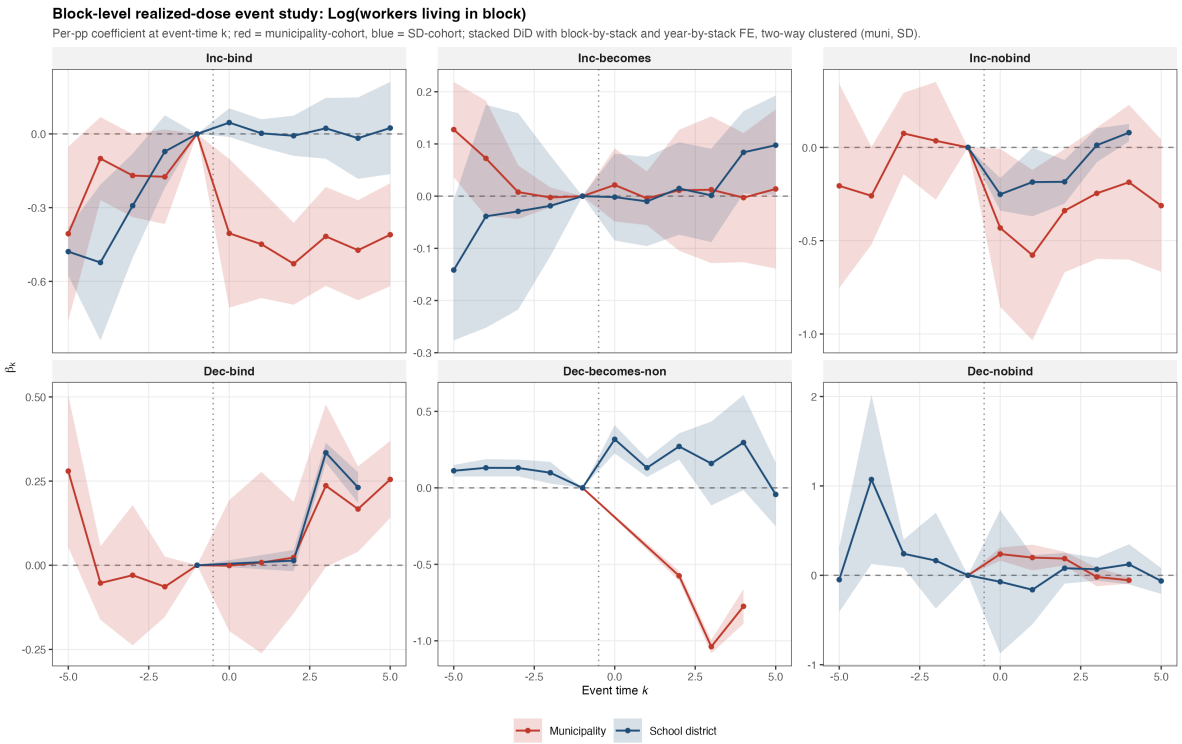
At both levels the substantive conclusion is the same: the Inc-bind cell is consistent with parallel trends under sensitivity bounds, the Inc-becomes and Inc-nobind placebo cells reflect cohort selection on pre-event trajectories rather than treatment-induced trends, and the post-event Inc-bind response is the cell the binding rule predicts.

Table 18: Realized-dose event study by binding-state cell — block level (muni cohort).

Cell	Log(workers live)	Log(workers work)	Share live+work in muni
Inc-bind (post)	-0.447	-0.124	-0.070
	(0.046)	(0.058)	(0.023)
Inc-bind (pre)	-0.213	0.004	0.022
	(0.060)	(0.042)	(0.012)
Inc-bind Wald p	0.0030	0.1427	<0.0001
Inc-becomes (post)	0.009	-0.028	-0.024
	(0.024)	(0.016)	(0.006)
Inc-becomes (pre)	0.051	0.040	0.012
	(0.020)	(0.016)	(0.003)
Inc-becomes Wald p	0.0031	0.1239	<0.0001
Inc-nobind (post)	-0.349	-1.468	0.008
	(0.081)	(0.070)	(0.007)
Inc-nobind (pre)	-0.089	-0.369	0.013
	(0.092)	(0.053)	(0.011)
Inc-nobind Wald p	0.0480	<0.0001	<0.0001
Dec-bind (post)	0.115	0.317	-0.031
	(0.040)	(0.148)	(0.010)
Dec-bind (pre)	0.033	0.378	0.013
	(0.043)	(0.114)	(0.011)
Dec-bind Wald p	<0.0001	<0.0001	<0.0001
Dec-becomes-non (post)	-0.796	-0.106	-0.008
	(0.021)	(0.018)	(0.007)
Dec-becomes-non (pre)	—	—	—
Dec-becomes-non Wald p	—	—	—
Dec-nobind (post)	0.110	-0.009	-0.016
	(0.021)	(0.027)	(0.003)
Dec-nobind (pre)	—	—	—

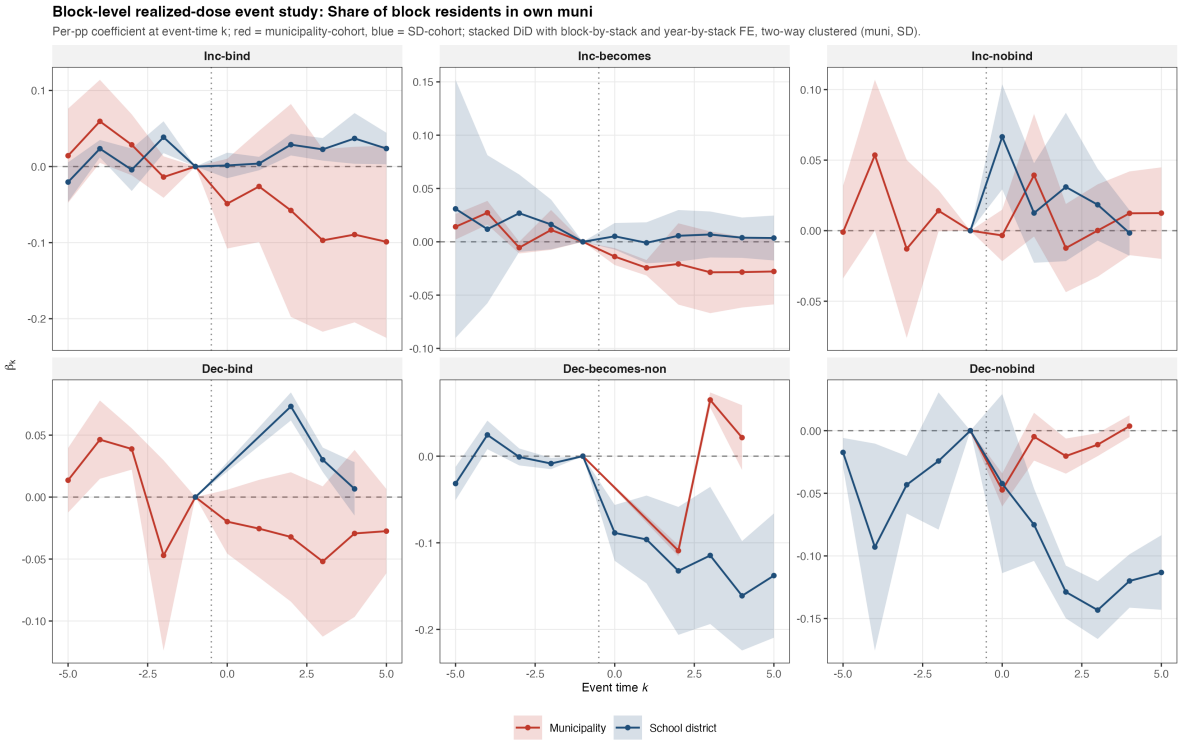
Figure 13 plots the per-event-time coefficients underlying that table, faceted by binding-state cell.

Figure 13: Block-level realized-dose event study on the log count of workers living in the block: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.



Notes: Per-pp realized-dose coefficient at each event-time, block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects, two-way clustered at the municipality and school-district levels. Red: the municipality-cohort stack (event time relative to first municipality-rate change). Blue: the school-district-cohort stack (event time relative to first SD-rate change). Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Figure 14: Block-level realized-dose event study on the share of block residents working in their own municipality: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.



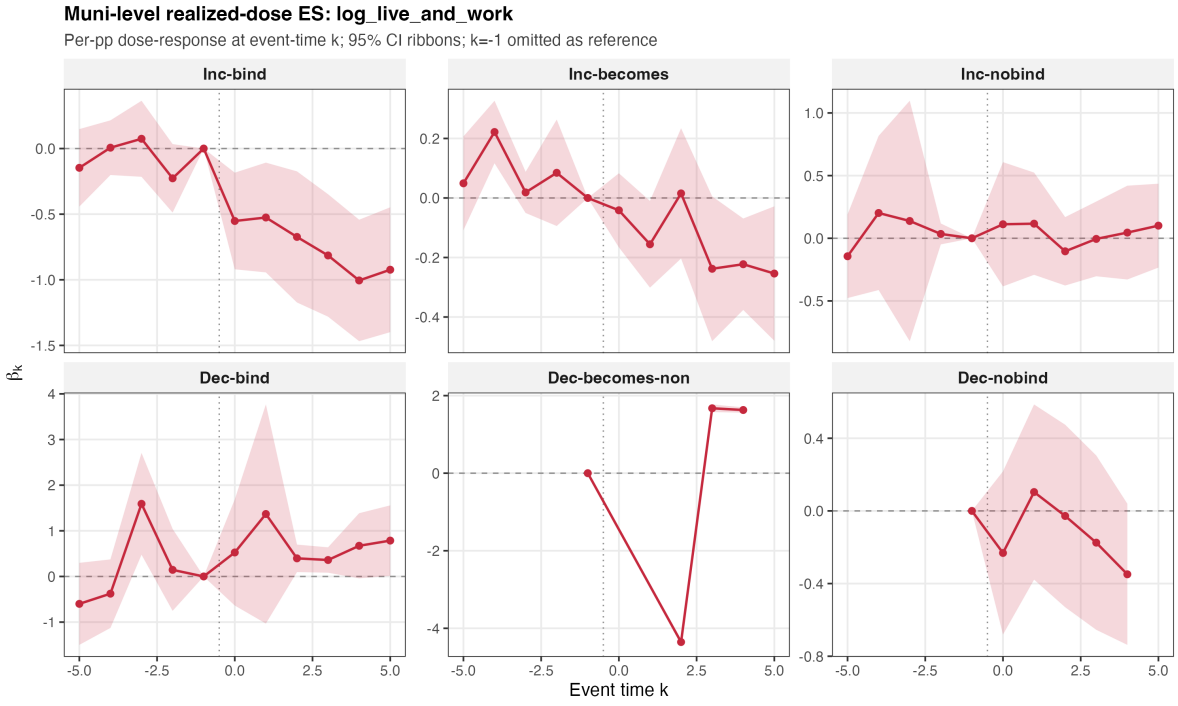
Notes: See Figure 13.

The Inc-becomes cell warrants a separate note. The headline stacked-DiD coefficient is essentially zero at the block level (−0.013), which initially looks like a failure of the predicted intermediate response. The realized-dose ES tells a different story: the pre-event lead is +0.080 ($|t| \approx 3$) and the post-event mean is +0.009, so the post-minus-pre difference is −0.071 per pp—about a third of the Inc-bind magnitude, in the predicted direction, but masked in the single-coefficient specification by selection on positive pre-event growth in Inc-becomes municipalities. The block-level Inc-becomes coefficient is therefore consistent with the partial-bill-change prediction of the institutional rule once the pre-trend is netted out, but I do not claim it as a clean causal estimate because the strong-parallel-trends assumption fails for this cell. At the municipality level the corresponding Inc-becomes coefficient is -0.181^{**} in the headline specification, and its realized-dose pre-trend is smaller (+0.055, $|t| \approx 1.3$);

the municipality-level signal is detectable in the post coefficient because the municipality-by-stack fixed effects absorb less of the within-municipality heterogeneity that attenuates the block-level coefficient. The block-level Inc-bind cell is the only combination of cell and estimating unit where the strong parallel-trends diagnostic passes cleanly, and is therefore the specification on which the paper rests its primary causal claim. The full event-time coefficient detail is in the Companion Appendix.

For completeness, the municipality-level post-event trajectory on Inc-bind log live-and-work is uniformly negative and individually significant from $k = +1$ onward (coefficients of -0.53, -0.67, -0.81, -1.01, -0.92 at $k = 1$ through $k = 5$, all $p < 0.05$), walking down monotonically across the post-window.

Figure 15: Municipality-level realized-dose event study on log(residents who live and work in the municipality), faceted by binding-state cell.



Notes: Per-pp realized-dose coefficient at each event-time, municipality-level stacked-continuous DiD with municipality-by-stack and year-by-stack fixed effects, cluster-robust by municipality. The Inc-bind facet shows pre-event coefficients mostly null pointwise (joint Wald F-test $p = 0.23$), followed by a monotonically negative post-event trajectory.

The patterns on the remaining residence-side outcomes and the school-district-level analogues mirror the live-and-work panel: Inc-bind shows visible pre-event drift consistent with high-dose post-industrial selection, while the within-design Inc-nobind placebo is approximately flat. The full per-outcome figure set is available in the replication archive.

6.3 Summary

The results follow a benchmark–decomposition–mechanism narrative. Section 6.1 reports the pooled Inc/Dec benchmark, which approximates a conventional local-tax design but mixes binding and non-binding events. Section 6.2.1 reports the preferred causal estimate: at the municipality level, a 1 pp resident-rate increase that binds reduces log live-and-work by -0.573^{***} with a clean realized-dose pre-trend and a null Inc-nobind within-design placebo. The block-level analogue is direction- confirming (-0.340^{***} on log home count) but has nonzero pre-trends, so I treat it as a companion rather than the causal headline. Section 6.2.3 reports the school-district aggregation as a diagnostic non-result. Robustness and pre-trend diagnostics are in Section 7; full SD tables, block share outcomes, and the binary Sun–Abraham IW event study are in the Appendix.

7 Robustness

This section summarizes the battery of robustness checks behind the headline dose-response; full tables, figures, and detailed prose are collected in Appendix C. The realized-dose event-study pre-trend diagnostic (Table 10) was discussed alongside the headline in Section 6.2.1.

7.1 Pre/Post Act 32 Decomposition

Splitting the headline cell-×-dose specification at the Act-32 enforcement break (cohort first event year $<$ 2012 vs $\geq$ 2012) shows the municipality-level binding-cell effect strengthens after consolidated employer withholding takes effect (-0.593^{**} pre vs -0.546^{***} post on log

residents working in own municipality); the block-level effect is roughly constant pre and post, so the finer-geography sorting response predates the Act-32 enforcement break (see Appendix C.1).

7.2 Act 47 Distressed-Municipality Exclusion

The Inc-bind cohort is mechanically concentrated in Act 47 distressed cities (Pennsylvania General Assembly, 1987), because Act 47 is the legal designation that lets a municipality exceed the 0.5% statutory municipality-EIT cap. Selection-on-distress is therefore a candidate confounder: an Act 47 designation typically follows a period of population decline, so the rate hike happens after the municipality is already losing residents and bundles with state oversight, debt restructuring, and service cuts. Of the 24 Inc-bind municipalities, 9 match the DCED Act 47 designation list (Braddock, Carbondale, Coatesville, Farrell, Franklin, Hazleton, Jeannette, Johnstown, West Hazleton). Table 19 re-estimates the municipality-level Inc-bind coefficient on the remaining 31 non-Act-47 Inc-bind municipalities.

Table 19: Act 47 robustness: dropping distressed Inc-bind municipalities from the cousub headline

Outcome	All 40 Inc-bind	Drop k Act 47
`log_live_and_work`	-0.411^{***} (0.119)	-0.330^{*} (0.183)
`log_in_commuters`	-0.171 (0.133)	-0.176 (0.215)
`log_total_residents`	-0.092^{***} (0.035)	-0.070 (0.058)
`share_live_in`	-0.048^{***} (0.011)	-0.038^{***} (0.013)

Notes: Muni-level Inc-bind coefficient on each residence-side outcome, with same FE / cluster / controls as the headline. Column 1 is the headline (all 40 Inc-bind municipalities). Column 2 drops 9 Act 47 distressed-designation municipalities from the Inc-bind cohort; the remaining 31 Inc-bind municipalities are mostly Home Rule and Boroughs. Cluster-robust SEs in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The point estimates are direction-preserving and similar in magnitude when Act 47 munis

are dropped: the live-and-work coefficient moves from −0.41 to −0.33, share-live-in from −0.048 to −0.038. Statistical significance falls because dropping 23% of the treated cohort roughly doubles the standard errors, but the share-live-in coefficient is still significant at 1% and live-and-work at 10%. The pattern is consistent with sample shrinkage rather than with the headline being driven entirely by Act 47 selection.

8 Discussion

The continuous stacked DiD design produces four substantive findings. The block-level model is the preferred causal estimate; the municipality-level model is the policy-jurisdiction magnitude calibration; the school-district-level aggregation is an aggregation diagnostic and is reported in the Companion Appendix.

The central finding is that sorting is identified through the binding channel. The Inc-bind cell—rate hikes that move the bill—reduces the log count of workers living in the block at the block level by approximately 19 percent per pp (−0.21 log points, SE = 0.054). The municipality-level analogue is larger (−0.57 log points on the log count of residents working in their own municipality, ≈ 33% per pp). The Inc-nobind placebo—rate hikes whose post-event home stack remains below the nonresident rate—is statistically indistinguishable from zero on the block-level count outcomes (−0.056 on the log count of workers living in the block, −0.113 on the log count of workers at the block) and on the municipality-level the log count of residents working in their own municipality (−0.057). Inc-becomes lies between the two. The pattern matches the institutional rule's prediction: the same statutory rate change can produce a strong, moderate, or null behavioral response depending solely on whether the rate moves the bill. This is direct evidence that workers respond to taxes they actually pay rather than to statutory rates per se.

Block-level identification is preferred not because the joint Wald F-test is cleaner—it is not—but because the block observes both rate-setting jurisdictions at their native geogra-

phy without a within-jurisdiction aggregation step. The two estimating units give opposite verdicts on the joint Wald: at the municipality level ($N \approx 198$K) the test does not reject parallel trends ($p = 0.33$), while at the block level ($N \approx 21$M) it does ($p = 0.003$). I read this as selection at finer geography that the muni-level aggregation washes out, not as hyper-precision. The municipality-level result is therefore the defensible per-pp causal quantity, and the block-level estimate is a direction-confirming companion.

The joint regression also separates the two rate-setting channels. The municipality resident-rate Inc-bind coefficient on resident outcomes is large and consistent at both the block and municipality levels. The SD-resident-rate cells, estimated jointly in the same regression, produce smaller and direction-mixed effects: $+0.177^{*}$ at the municipality level on the log count of residents working in their own municipality, $+0.097$ on block-level log count of workers living in the block ($p = 0.12$), and $+0.066^{*}$ at the SD level (sign-reversed). The SD rate is not the dominant driver of residential sorting through the binding channel. This is a substantive change from an earlier draft that found a positive SD-rate "attraction" effect, which was an artifact of tract-centroid LODES aggregation; under block-level LODES with the unified two-rate-setter design, SD-rate effects on residents are small or wrong-signed.

The mechanism behind the workplace response appears to be local-services demand spillover rather than direct tax incidence on jobs. Industry-decomposed outcomes at the municipality level show consumer-facing jobs (retail, food, arts, other services) fall by approximately 26 percent in the Inc-bind cell (-0.30 log points, SE $= 0.15$) while tradeable jobs (manufacturing, wholesale, finance, professional services) are insulated (point estimate $+0.24$, statistically indistinguishable from zero). The composition shift is captured by the share of tradeable jobs rising approximately 11 percentage points in the binding cell. The decline in total jobs operates via local-services demand spillover from departing residents, not via direct nonresident-tax incidence.

The treated cohort is also disproportionately concentrated in Act 47 distressed cities, which raises a selection-on-distress concern: Act 47 designation typically follows a period of

decline and bundles rate-hike authority with state oversight, debt restructuring, and service cuts. Section 7.2 re-estimates the headline excluding the 9 Act 47 Inc-bind municipalities; the point estimates are direction-preserving and similar in magnitude (live-and-work moves from -0.41 to -0.33, share-live-in from -0.048 to -0.038), with standard errors roughly doubling as expected from a 23% sample reduction. The binding-state mechanism is therefore not a relabelling of Act 47 fiscal distress, though the municipality-level coefficient is identified off a treated set that includes substantial Act 47 weight.

Eight limitations merit acknowledgment. First, while the block-level stacked headline on the count outcomes is large, negative, and well-identified, the block-level share outcome (the share of block residents working in their own municipality) is null in the binding cell (-0.035, $p = 0.26$) and marginally significant in the Inc-nobind placebo (-0.037, $p = 0.015$). The share outcome at block scale appears to reflect mechanical denominator dynamics on very small populations rather than binding-state-specific behavior, so the share-based block evidence should be read as suggestive rather than identified. The count-based block evidence remains the cleanest test. Second, the school-district aggregation produces direction-mixed coefficients and a marginally non-zero realized-dose pre-event lead; the SD-level results should be read as confirming the directional sign of the municipality-rate effect under a coarser aggregation rather than as standalone identification of an SD-channel response. Third, the LODES OD data covers PA municipalities but not cross-state inflows in the intra-state main file used here; the auxiliary file (cross-state inflows) is downloaded but not yet incorporated, which would tighten the estimates of in-commuters for border municipalities.

Fourth, the paper estimates the residential-sorting response to the home-side resident rate but does not separately estimate the work-side response to the municipality nonresident rate (τ^{nr}). The nonresident rate enters the design only through the binding-state classification of cells. A complementary work-side specification—estimating in-commuter and total-jobs responses to τ^{nr} events at the municipality level, with a parallel binding-state decomposition—would test the workplace-side response separately. I defer this to a separate

paper.

Fifth, the design treats each treated municipality's stack as if its identifying variation were independent of the rate trajectories of its geographic neighbors. Pennsylvania municipalities are small and dense; a worker who moves out of a rate-raising municipality often relocates to an adjacent municipality whose own rate is also moving in some year of the panel. The headline specification absorbs this through never-changed-only controls and municipality-by-stack fixed effects, but it does not formally model spatial spillovers, so the per-pp coefficient is best read as a local displacement elasticity rather than a net-migration one. The **butts2023DiDSpatialSpillovers** estimator generalizes stacked DiD to allow for SUTVA violations from spatial spillovers across treated units, using an explicit spatial weighting matrix on the adjacency graph. Applying it to the PA cousub adjacency graph is left for future work.

Sixth, I do not currently estimate the headline using the heterogeneity-robust continuous-treatment difference-in-differences estimator of **deChaisemartin2022Continuous**, which generalizes the de Chaisemartin–D'Haultfœuille framework (**deChaisemartin2020TWFE**; **deChaisemartin2023ImperfectPT**) to continuously distributed treatments at every period and leverages switchers (units that move both up and down in dose) for identification. An earlier iteration of this paper used a prototype CDH-continuous specification at the cousub level; the block-level analog is computationally prohibitive on the current stacked panel ($\sim$ 21M block-years $\times$ multiple cohorts) and is left for future work. Re-estimating the headline with `did_multiplegt_dyn` (**deChaisemartin2025MultipleGtDyn**)—once the package documentation and switcher-based identification at this scale are stable—would also let me exploit the rate-decrease (Dec) cohorts, currently too thin under conventional cluster-robust inference, as identification rather than just as a directional mirror.

Seventh, LODES is a stock dataset rather than a person-linked flow dataset. For each (residence block, workplace block, year), I observe a count of workers; I do not observe individual migration histories. The headline coefficient on the log count of workers living in the

block is therefore a net elasticity—the combined effect of move-outs and move-ins—rather than separate estimates of the gross outflow and gross inflow margins. The paper's evidence that municipal rate hikes reduce the residential count is consistent both with workers leaving and with workers not arriving to replace those who left; LODES alone cannot decompose the two. Decomposing into gross outflow and gross inflow would require person-linked administrative data: the IRS Statistics of Income individual master file, the American Community Survey microdata with full residence-history linkage, or—most directly—the LEHD Job-to-Job Flows file, which tracks the same workers across employer transitions at quarterly frequency. All three are restricted-access products requiring approved-project status under the Census Bureau's Federal Statistical Research Data Centers (FSRDCs) or IRS Statistics of Income agreements; the microdata cannot be exported and must be analyzed at a physical FSRDC location. Replicating the present analysis on person-linked data is a clear extension but a substantially larger undertaking with its own access timeline.

Eighth, the paper does not estimate a comparable elasticity for the property tax, the other major local-revenue lever in Pennsylvania. The Oates (1969) capitalization theory predicts a substantially smaller residential-sorting elasticity for property tax than for the EIT, because property tax falls on owners and capitalizes into house prices rather than on workers directly. Estimating a comparable property-tax elasticity would test the falsification implication "is the EIT effect specifically about the EIT, or any local tax raising?" but requires a 2002–2019 municipality-level millage panel that DCED does not publish (the DCED Annual Financial Report schema dropped per-municipality millage in 2006). Constructing this panel from county assessor records is left for future work.

9 Conclusion

When do local income taxes cause residential sorting? The institutional answer in Pennsylvania is that they cause sorting when the piecewise max-rule makes the statutory rate increase

change the worker's actual bill. At the census block, the level at which both rate-setting jurisdictions enter the regression at their native geography, a 1 percentage-point municipal resident-rate increase in the Inc-bind cell reduces the log count of workers living in the block by roughly -0.21 log points, approximately 19%, with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.213$, Wald $p = 0.003$). The within-design Inc-nobind placebo, where the same statutory increase does not move the bill, is statistically null on the same count outcome. The response is therefore tied to tax liability rather than to the statutory rate change itself, and the binding-state decomposition is what distinguishes the two.

Municipality-level estimates are larger (-0.57 log points on the log count of residents working in their own municipality) but pre-trended, and they enter the paper as policy-jurisdiction magnitude calibration rather than as a second causal headline. School-district aggregations do not robustly detect a sorting response and are reported as a diagnostic non-result. Industry decomposition shows that consumer-facing jobs fall in the binding cell while tradeable jobs are insulated, suggestive of a local-services demand spillover from departing residents rather than direct nonresident-tax incidence on jobs.

The narrow, credible claim is that local income taxes cause residential sorting when they bind. The same design framework—a continuous stacked DiD with binding-state cell decomposition, a realized-dose event study as a strong-parallel-trends diagnostic, and a within-design placebo cell whose theoretical prediction is zero—is portable to other settings where a piecewise tax rule generates predictable cross-sectional variation in whether a statutory rate change moves the worker's bill. Local sales taxes with use-tax-credit rules, county-level income-tax credits across state borders, and special tax districts with overlapping jurisdictions are candidate extensions.

References

Agrawal, D. R. (2015). The tax gradient: Spatial aspects of fiscal competition. *American Economic Journal: Economic Policy*, *7*(2), 1–29. https://doi.org/10.1257/pol.20120360

Agrawal, D. R. (2026). Lessons on State and Local Income Taxes from the Twenty-first Century and Challenges for the Future. *CESifo Working Papers*.

Agrawal, D. R., & Foremny, D. (2019). Relocation of the Rich: Migration in Response to Top Tax Rate Changes from Spanish Reforms. *The Review of Economics and Statistics*, *101*(2), 214–232. https://doi.org/10.1162/rest_a_00764

Bagchi, S. (2022). The Effects of Introducing Withholding on Tax Compliance: Evidence from Pennsylvania's Local Earned Income Tax. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4291598

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230. https://doi.org/10.1016/j.jeconom.2020.12.001

de Chaisemartin, C., D'Haultfœuille, X., & Vazquez-Bare, G. (2024). Difference-in-difference estimators with continuous treatments and no stayers. *AEA Papers and Proceedings*, *114*, 393–397. https://doi.org/10.1257/pandp.20241049

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277. https://doi.org/10.1016/j.jeconom.2021.03.014

Graham, M. R., Kutzbach, M. J., & McKenzie, B. (n.d.). DESIGN COMPARISON OF LODES AND ACS COMMUTING DATA PRODUCTS.

Pennsylvania Department of Community and Economic Development. (2024a). MunStats portal: Municipal Statistics and Local Government Data.

Pennsylvania Department of Community and Economic Development. (2024b). Statewide Municipal Annual Financial Report (AFR) Database.

Pennsylvania General Assembly. (1932). Sterling Act, Act of August 5, 1932, P.L. 45.

Pennsylvania General Assembly. (1965). Local Tax Enabling Act, Act 511 of 1965.

Pennsylvania General Assembly. (1972). Home Rule Charter and Optional Plans Law.

Pennsylvania General Assembly. (1987). Municipalities Financial Recovery Act, Act 47 of 1987.

Pennsylvania General Assembly. (2006). Taxpayer Relief Act, Act 1 of 2006 (Special Session).

Tiebout, C. M. (1956). A pure theory of local expenditures. *Journal of Political Economy*, *64*(5), 416–424. https://doi.org/10.1086/257839

U.S. Census Bureau. (2024a). Annual Survey of School System Finances (F-33).

U.S. Census Bureau. (2024b). LEHD Origin-Destination Employment Statistics (LODES), Version 8.

Yang, L. (, & Heim, B. T. (2017). RESPONSIVENESS OF INCOME TO LOCAL INCOME TAXES: EVIDENCE FROM INDIANA. *National Tax Journal*, *70*(2), 367–391. https://doi.org/10.17310/ntj.2017.2.05

Young, C., Varner, C., Lurie, I. Z., & Prisinzano, R. (2016). Millionaire migration and taxation of the elite: Evidence from administrative data. *American Sociological Review*, *81*(3), 421–446. https://doi.org/10.1177/0003122416639625

A Full Decomposition Tables

A.1 Block-level share outcomes

Block-level share outcomes from the same joint regression, reported for completeness; the share outcomes are mechanically fragile at block geography (see body discussion).

Table 20: Block-level dose-response: muni resident-rate cells (share outcomes, supporting)

Outcome	Share live+work in muni
Inc-bind	-0.062
	(0.053)
Inc-becomes	-0.035***
	(0.013)
Inc-nobind	-0.162***
	(0.030)
Dec-bind	-0.031
	(0.037)
Dec-becomes-non	-0.211***
	(0.032)
Dec-nobind	-0.009***
	(0.002)
N	20,050,835
R^2	0.438
Within R^2	0.000
Dep. mean	0.103
Dep. SD	0.158

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Block-level share outcomes are reported for completeness. Sparse denominators and endogenous denominator dynamics make these estimates mechanically fragile at block geography; they are not used to identify the headline causal effect. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 21: Block-level dose-response: SD resident-rate cells (share outcomes, supporting)

Outcome	Share live+work in muni
Inc-bind	0.008***
	(0.003)
Inc-becomes	-0.001
	(0.006)
Dec-becomes-non	-0.084***
	(0.011)
Dec-nobind	-0.155***
	(0.027)
N	20,050,835
R^2	0.438
Within R^2	0.000
Dep. mean	0.103
Dep. SD	0.158

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Block-level share outcomes are reported for completeness; not used for headline identification. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind, Dec-bind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.2 Municipality-level: Dec cells + full specification

Table 22: Headline stacked-continuous DiD: municipality-rate Dec cells.

Outcome	Dec-bind (n=11)	Dec-becomes-non (n=4)	Dec-nobind (n=15)
log(live_and_work)	+0.241*** (0.078)	−0.260 (0.272)	+0.332** (0.165)
log(in_commuters)	+0.052 (0.080)	+0.496 (0.339)	+0.013 (0.121)
log(out_commuters)	−0.068 (0.061)	−0.628 (0.416)	+0.132 (0.101)
log(total_jobs)	+0.111 (0.073)	+0.420 (0.265)	+0.022 (0.116)
log(total_residents)	−0.056 (0.043)	−0.651 (0.420)	+0.135 (0.099)
share_live_in	+0.054* (0.030)	−0.071 (0.095)	+0.012 (0.030)

Notes: Same regression as the municipality-level headline decomposition. Cluster-robust SEs by municipality. Dec cells are thin; not used to draw causal conclusions individually. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 23: Full municipality-level specification on log(live_and_work): all cells + controls.

Term	Coefficient	SE
Muni resident-rate cells (vs. NC^{muni})		
Inc-bind $\times\Delta\tau^{\text{muni}}$	-0.573^{***}	0.096
Inc-becomes $\times\Delta\tau^{\text{muni}}$	-0.181^{**}	0.072
Inc-nobind $\times\Delta\tau^{\text{muni}}$	-0.062	0.169
Dec-bind $\times\Delta\tau^{\text{muni}}$	$+0.241^{***}$	0.078
Dec-becomes-non $\times\Delta\tau^{\text{muni}}$	-0.260	0.272
Dec-nobind $\times\Delta\tau^{\text{muni}}$	$+0.332^{**}$	0.165
School district rate cells (vs. NC^{SD})		
Inc-bind $\times\Delta\tau^{\text{SD}}$	$+0.177^{*}$	0.099
Inc-becomes $\times\Delta\tau^{\text{SD}}$	$+0.103$	0.090
Dec-bind $\times\Delta\tau^{\text{SD}}$	-0.721^{***}	0.191
Dec-becomes-non $\times\Delta\tau^{\text{SD}}$	-0.223^{**}	0.109
Dec-nobind $\times\Delta\tau^{\text{SD}}$	$+0.048$	0.179
Share-based controls		
share_police	$+0.097$	0.100
share_streets	$+0.049$	0.039
share_debt_svc	-0.058	0.051
share_eit	-0.022	0.085
share_re	-0.182^{**}	0.087
share_intergov	$+0.016$	0.044
N	218,685	
R^2	0.964	
Within R^2	0.0007	
Dep. mean	3.677	
Dep. SD	1.934	
FE	stack-municipality + stack-year	
Clusters	1,945 munis	

Notes: All coefficients from a single regression on log(live_and_work). Reference category on each rate-setter is NC. Cluster-robust SEs by municipality. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Realized-Dose Event Studies, Full Set

Per-cell realized-dose Sun–Abraham event studies at each estimating unit, on each residence-side outcome. The block-level log(home_count) figure and the municipality-level log(live_and_work) figure appear in the body (§6.2.4); the remaining outcomes and the SD-level analogues are here.

B.1 Municipality-level, by cell

Figure 16: Muni-level realized-dose ES on log(in-commuters), by cell.

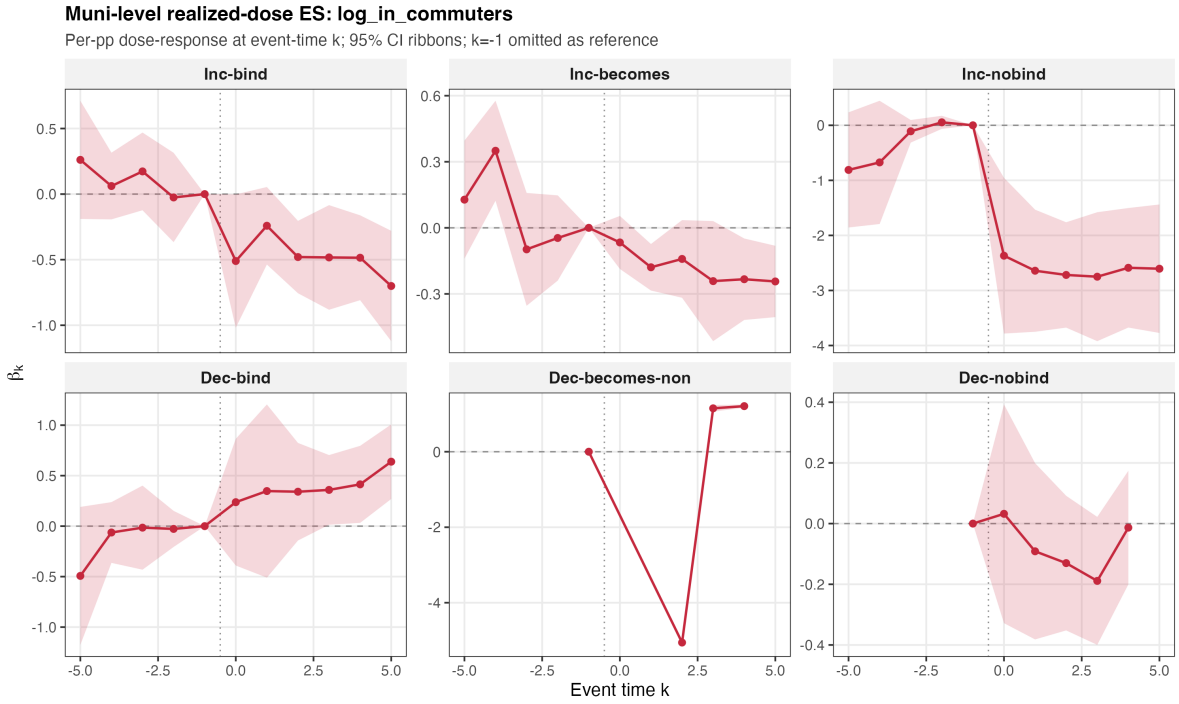


Figure 17: Muni-level realized-dose ES on log(total residents), by cell.

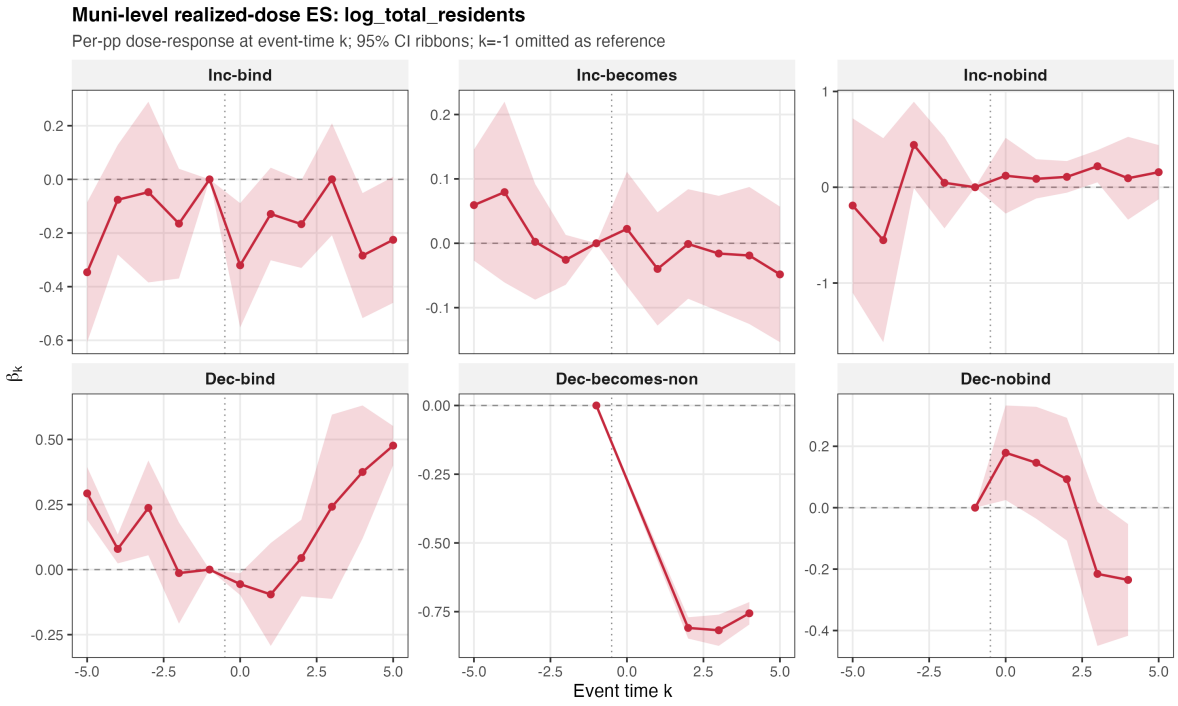
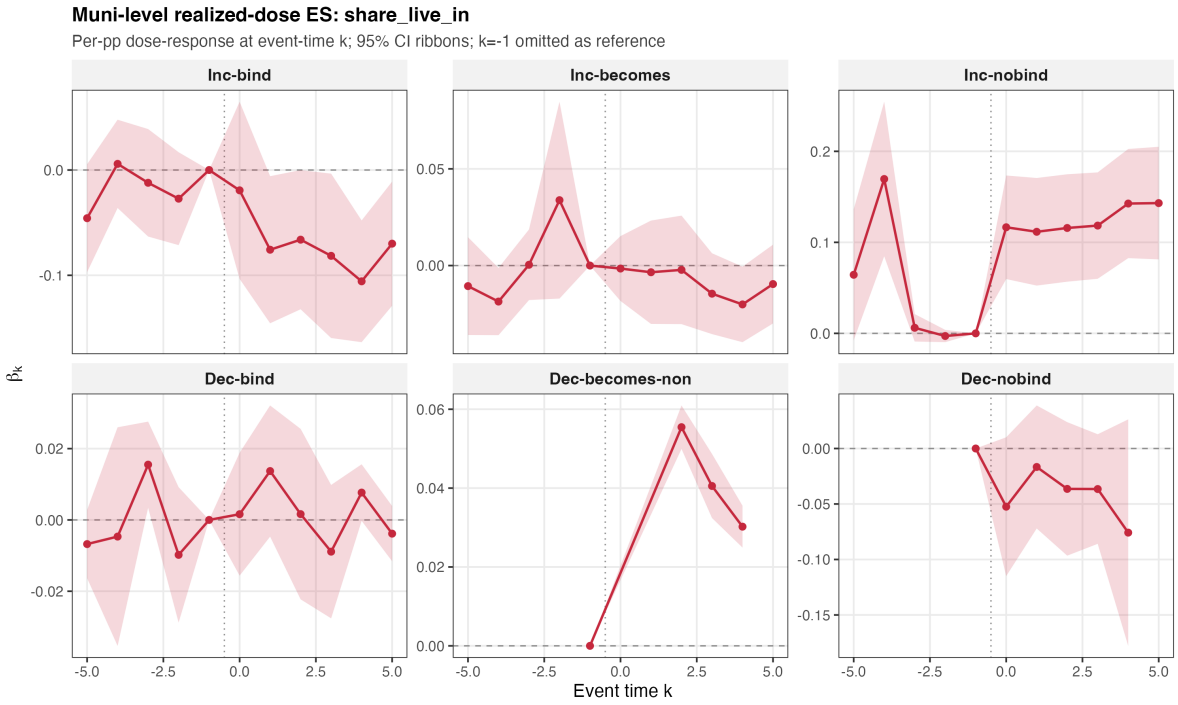


Figure 18: Muni-level realized-dose ES on the share of workers living in their own municipality, by cell.



B.2 School-district-level, by cell

Figure 19: SD-level realized-dose ES on log(live_and_work), by cell.

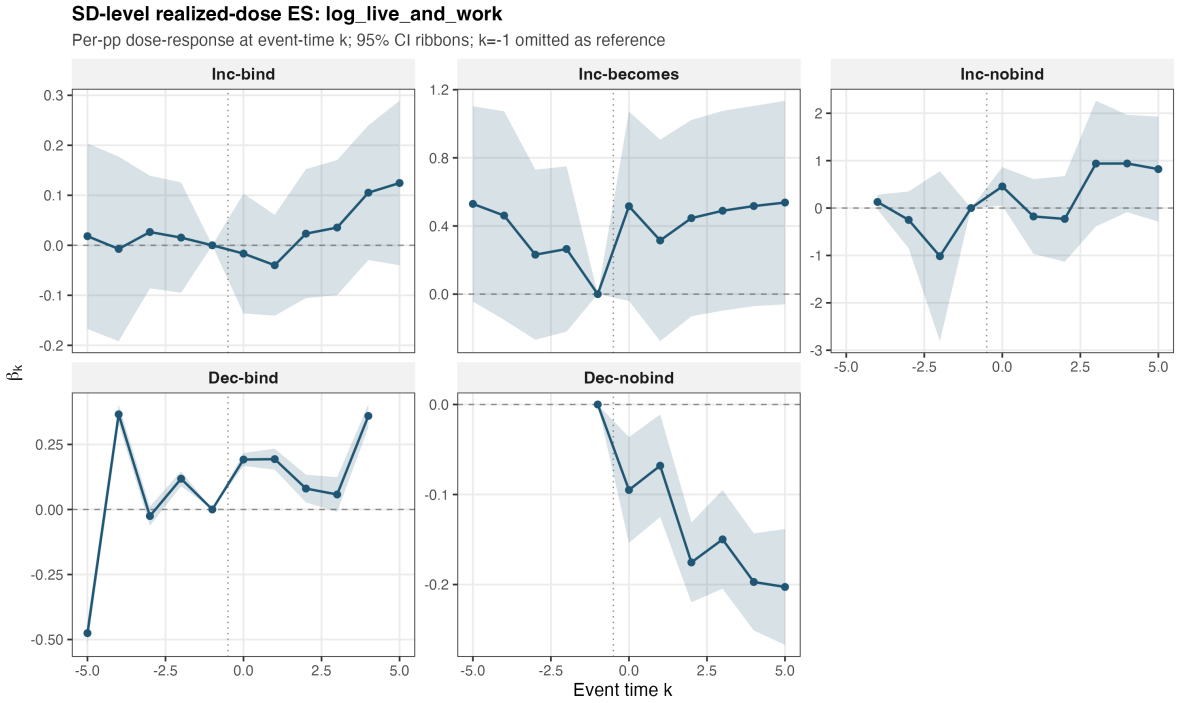


Figure 20: SD-level realized-dose ES on log(in-commuters), by cell.

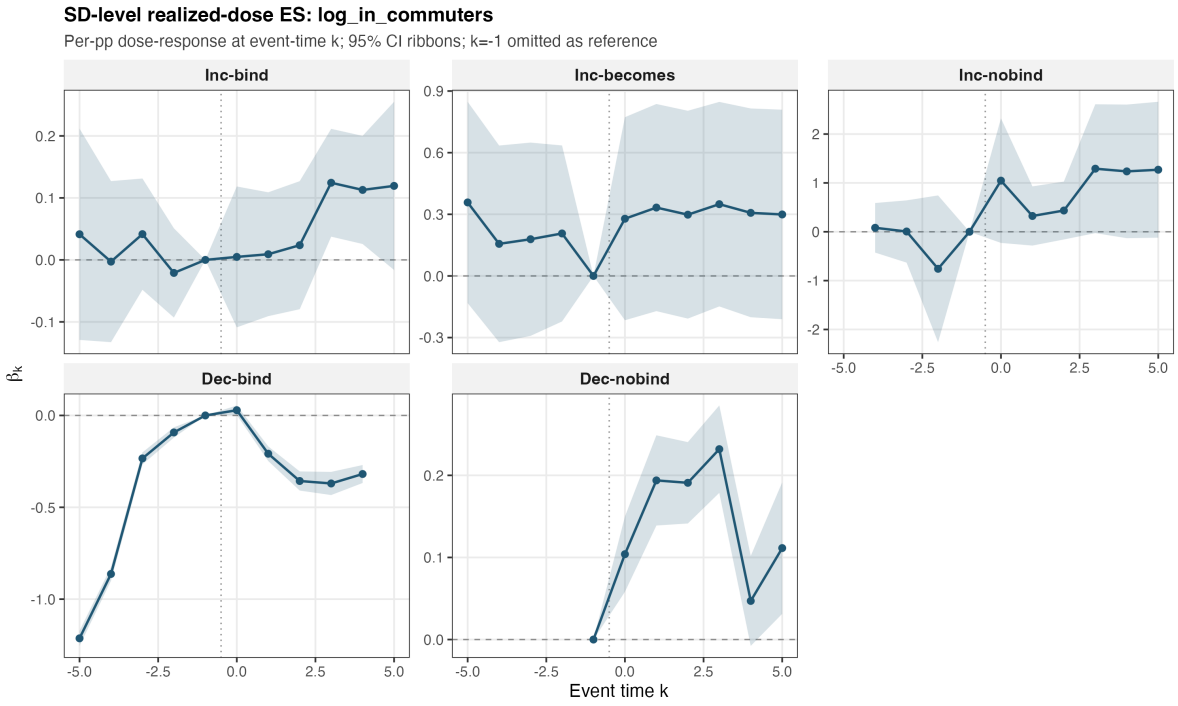


Figure 21: SD-level realized-dose ES on log(total residents), by cell.

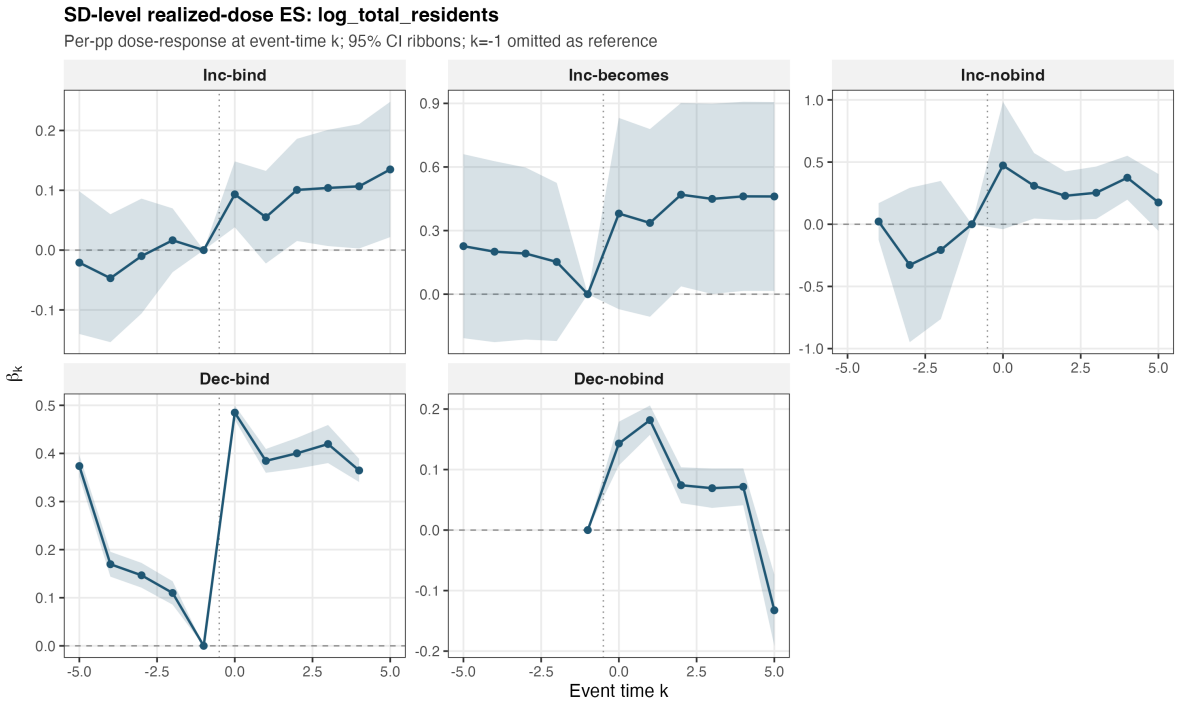
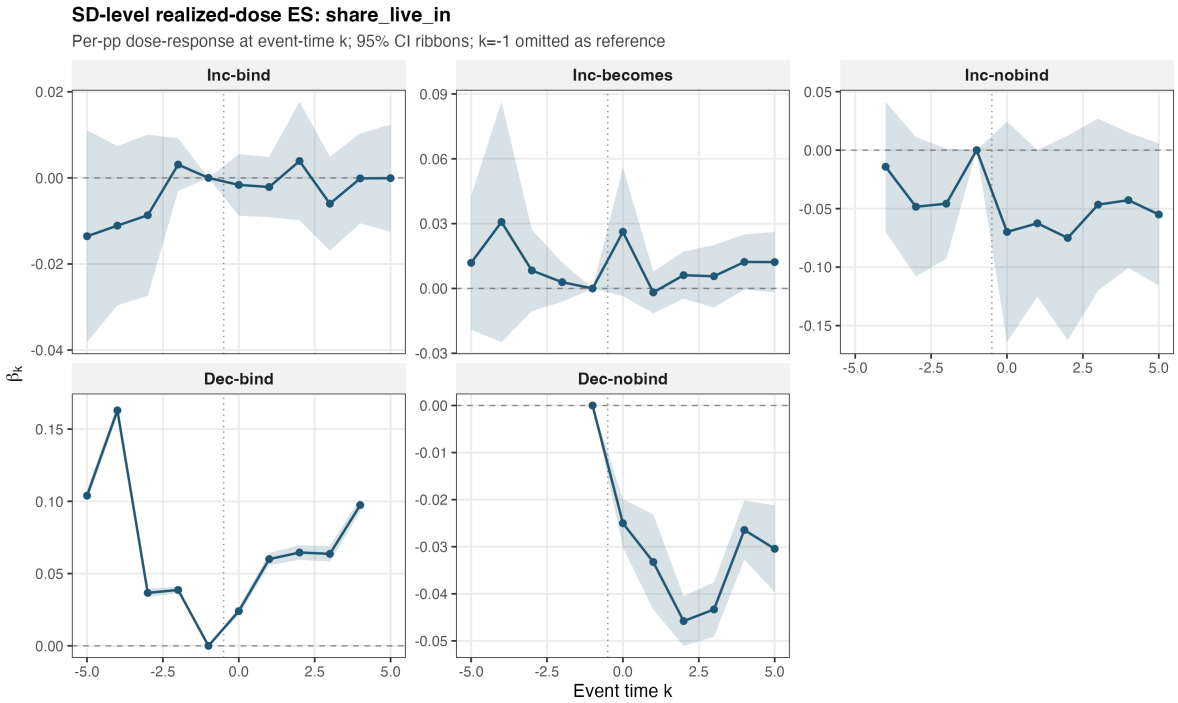


Figure 22: SD-level realized-dose ES on the share of workers living in their own municipality, by cell.



B.3 Block-level, by cell

Figure 23: Block-level realized-dose ES on log(work count) (in-block jobs), faceted by binding-state cell; municipality-cohort (red) and SD-cohort (blue).

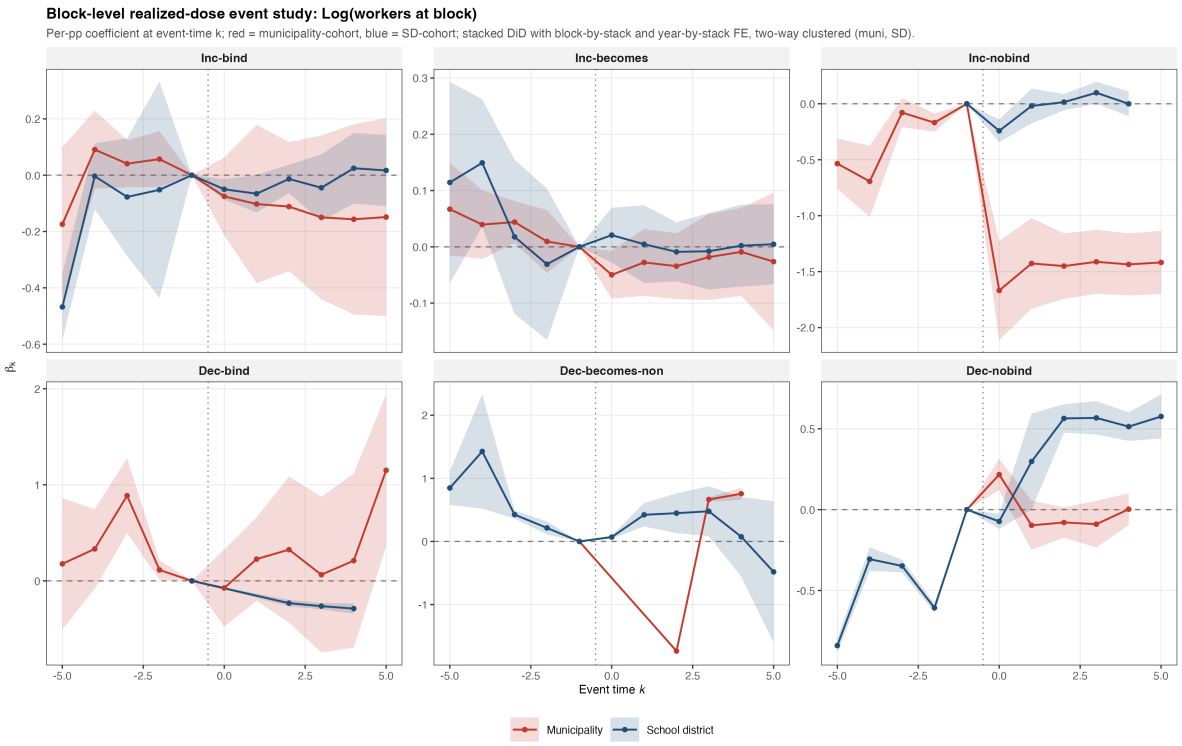
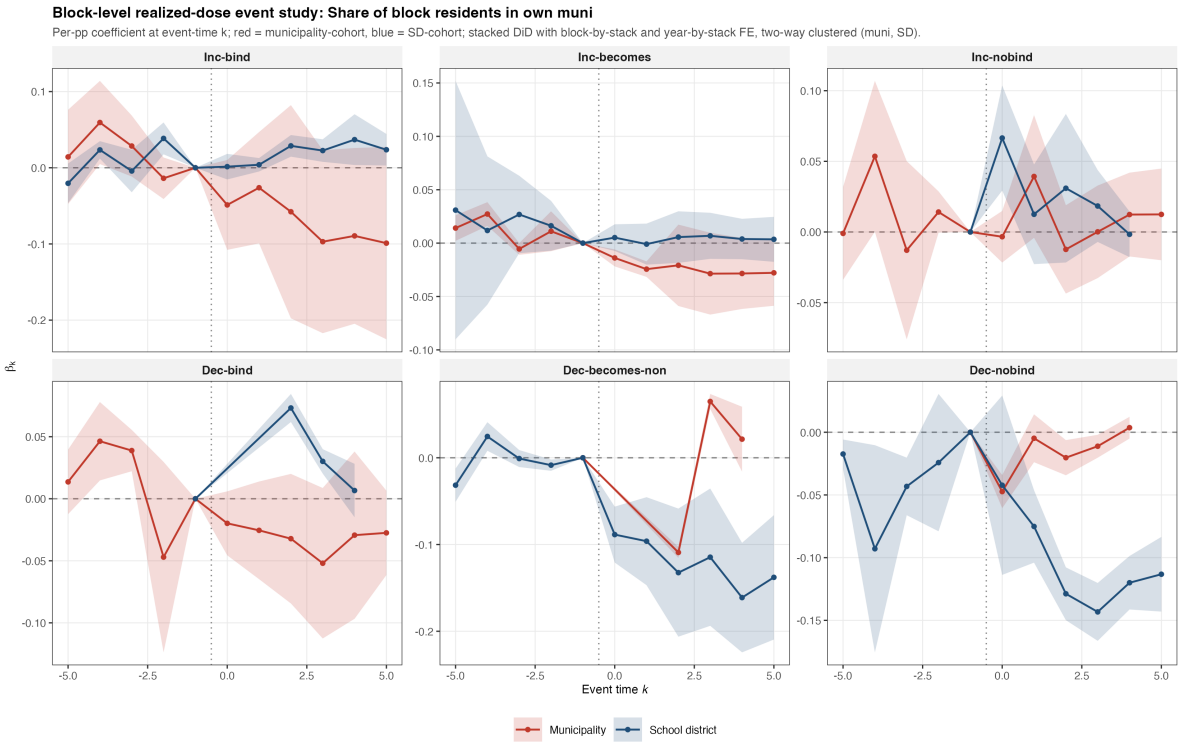


Figure 24: Block-level realized-dose ES on the share of block residents working in their own municipality, faceted by binding-state cell.



C Robustness

Full tables behind the robustness summaries in Section 7.

C.1 Pre/post Act 32 split

Headline cell × dose specification re-estimated on pre-Act-32 ($c < 2012$) and post-Act-32 ($c \geq 2012$) cohort subsamples at each estimating unit.

Table 24: Pre-/post-Act-32 split — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Inc-bind				
Pre-Act 32	-0.593**	-0.412**	-0.125**	-0.068**
	(0.235)	(0.181)	(0.051)	(0.030)
Post-Act 32	-0.546***	-0.443***	-0.073	-0.041***
	(0.168)	(0.149)	(0.085)	(0.005)
Inc-becomes				
Pre-Act 32	-0.317***	-0.289**	-0.102**	-0.042*
	(0.121)	(0.117)	(0.040)	(0.023)
Post-Act 32	-0.116	-0.136	-0.000	0.004
	(0.109)	(0.112)	(0.017)	(0.009)
Inc-nobind				
Post-Act 32	-0.091	-0.302	0.366***	-0.011
	(0.113)	(0.370)	(0.095)	(0.013)
Dec-bind				
Pre-Act 32	0.373***	0.255***	0.277***	0.012***
	(0.024)	(0.024)	(0.010)	(0.002)
Post-Act 32	0.063	0.043	-0.137	0.008***
	(0.145)	(0.186)	(0.101)	(0.003)
Dec-becomes-non				
Pre-Act 32	-1.211***	0.193***	-1.478***	-0.145***
	(0.070)	(0.072)	(0.033)	(0.005)
Dec-nobind				
Post-Act 32	—	0.149***	-0.228***	0.013***
		(0.010)	(0.005)	(0.001)
Pre-Act 32				
N	193,507	211,564	214,931	211,573
R^2	0.964	0.959	0.981	0.825
Within R^2	0.001	0.001	0.001	0.001
Dep. mean	3.661	5.838	6.688	0.093
Dep. SD	1.919	1.877	1.294	0.075
Post-Act 32				
N	193,285	211,338	214,716	211,347
R^2	0.964	0.959	0.981	0.825
Within R^2	0.001	0.001	0.001	0.001
Dep. mean	3.659	5.836	6.686	0.093
Dep. SD	1.919	1.877	1.294	0.075

Table 25: Pre-/post-Act-32 split — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni
Inc-bind				
Pre-Act 32	0.027	0.057	0.106***	0.003
	(0.038)	(0.046)	(0.030)	(0.004)
Inc-becomes				
Pre-Act 32	0.290*	0.155	0.272**	0.008
	(0.162)	(0.144)	(0.128)	(0.006)
Post-Act 32	0.043	0.523***	0.866***	-0.052
	(0.607)	(0.112)	(0.278)	(0.054)
Inc-nobind				
Pre-Act 32	0.794	1.149*	0.421***	-0.035
	(0.506)	(0.633)	(0.148)	(0.023)
Dec-bind				
Post-Act 32	0.056	0.049	0.178***	-0.000
	(0.075)	(0.111)	(0.044)	(0.004)
Dec-nobind				
Pre-Act 32	-0.126***	0.155***	0.076***	-0.032***
	(0.025)	(0.028)	(0.016)	(0.002)
Pre-Act 32				
N	51,382	51,416	51,416	51,416
R^2	0.986	0.987	0.987	0.953
Within R^2	0.004	0.003	0.004	0.004
Dep. mean	6.261	8.335	8.769	0.125
Dep. SD	1.298	1.202	0.909	0.068
Post-Act 32				
N	50,947	50,981	50,981	50,981
R^2	0.986	0.987	0.987	0.953
Within R^2	0.003	0.001	0.003	0.004
Dep. mean	6.262	8.335	8.770	0.125
Dep. SD	1.297	1.203	0.908	0.068

Notes: Cluster-robust standard errors in parentheses. Treated cohorts split at first event year $c < 2012$ (Pre-Act 32) vs $c \geq 2012$ (Post-Act 32). Cells omitted (no estimable coefficients): Dec-becomes-non. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 26: Pre-/post-Act-32 split — block level

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni
Inc-bind			
Pre-Act 32	-0.355***	-0.166*	-0.075
	(0.051)	(0.094)	(0.058)
Post-Act 32	-0.093	0.199	-0.006
	(0.158)	(0.204)	(0.013)
Inc-becomes			
Pre-Act 32	-0.123***	-0.113***	-0.017
	(0.016)	(0.026)	(0.019)
Post-Act 32	0.008	-0.041***	-0.042***
	(0.010)	(0.010)	(0.010)
Inc-nobind			
Post-Act 32	0.659***	0.104**	-0.146***
	(0.085)	(0.052)	(0.036)
Dec-bind			
Pre-Act 32	0.172***	0.820***	0.028***
	(0.020)	(0.016)	(0.003)
Post-Act 32	0.049	-0.445***	-0.072***
	(0.095)	(0.130)	(0.015)
Dec-becomes-non			
Pre-Act 32	-1.013***	0.853***	-0.180***
	(0.171)	(0.196)	(0.027)
Dec-nobind			
Post-Act 32	-0.200***	0.270***	-0.009***
	(0.004)	(0.005)	(0.002)
Pre-Act 32			
N	8,339,642	3,083,788	8,339,642
R^2	0.856	0.867	0.469
Within R^2	0.000	0.000	0.000
Dep. mean	2.209	2.268	0.105

C.2 Sample variants

Three sample variants for the DCED nonresident-rate panel (strict, filled, drop-incomplete) leave the headline municipality Inc-bind coefficient on log live-and-work essentially unchanged at -0.573^{***}.